The Risk and Return of Cryptocurrency Carry Trade*

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Abstract

This paper analyzes the risk and return dynamics associated with cryptocurrency carry trade. A cross-sectional carry trade strategy, involving the purchase of high-interest cryptocurrencies and shorting of low-interest cryptocurrencies, yields an annualized return of 43.4% with a Sharpe ratio of 0.74. Our results show that the carry returns cannot be explained by prevailing cryptocurrency factors, including market, size, momentum, volatility, liquidity, downside risk, and platform collapse risks. Additionally, our analysis does not find any substantial connection between these returns and the fiat currency carry trade. Instead, our findings suggest that a significant portion of cryptocurrency carry trade returns can be attributed to a premium for equity market volatility risk. This study highlights the inter-asset class linkages between equity risk factors and cryptocurrency returns.

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1 Introduction

Cryptocurrency has emerged as a new investment vehicle in recent years, offering novel opportunities to investors. This study examines the performance of the cryptocurrency carry trade. Previous research on fiat currencies shows that the carry trade can generate positive returns for investors by leveraging the differential in interest rates between low- and high-yielding currencies.[1] The study by Koijen, Moskowitz, Pedersen and Vrugt (2018) broadens the scope of the carry trade concept to other asset classes and finds that this strategy delivers positive returns in global equity, commodity, bond, options, and other markets.

While the carry trade in fiat currencies is a well-studied trading strategy, much less is known about its dynamics in the realm of digital currencies. In fact, many cryptocurrency platforms, such as Bitfinex, offer loanable programs that enable investors to earn interest by depositing their coins or borrowing coins at interest for margin trading. These mechanisms resemble the traditional carry trade in fiat markets but come with the unique risk and return inherent to cryptocurrencies. Unlike fiat currencies, where returns are closely tied to macroeconomic fundamentals, as shown by Lustig and Verdelhan (2007, 2011) and Brunnermeier, Nagel and Pedersen (2008), cryptocurrency valuations are less influenced by macroeconomic factors and monetary policies due to their decentralized nature. This suggests that the risk-return dynamics of the cryptocurrency carry trade may diverge from those of fiat currencies, offering new investment opportunities and broader insights into how cryptocurrencies differ from traditional assets.

By obtaining the lending and borrowing rates from Bitfinex, we sort cryptocurrencies into portfolios based on their interest rates, following the common practice of the fiat currency carry trade. Given the limited number of loanable cryptocurrencies, we divide the tokens into three groups of equal size. The High portfolio contains tokens with the highest deposit rates, while the Low portfolio contains those with the lowest deposit rates. The portfolios are rebalanced weekly, in line with Liu, Tsyvinski and Wu (2022) and Cong, He and Tang (2023). We find a clear, monotonically increasing relationship between the average excess returns of cryptocurrencies with their interest rates across the sorted portfolios from the perspective of a U.S. investor. Specifically,

[1] The literature includes, but is not limited to, Burnside, Eichenbaum, Kleshchelski and Rebelo (2011); Lustig, Roussanov and Verdelhan (2011); Menkhoff, Sarno, Schmeling and Schrimpf (2012); Lettau, Maggiori and Weber (2014); Bekaert and Panayotov (2020); Cui, Filippou and Liu (2022).

high-interest-rate cryptocurrencies deliver an average annualized excess return of 93.9%, whereas low-interest-rate ones generate 50.4%. The carry trade strategy, which shorts low-interest-rate currencies and longs high-interest-rate ones, generates a decent performance with an average return of approximately 43.4% per year.

The performance is particularly noteworthy given the relatively lower level of risk compared to the aggregate cryptocurrency market. The carry trade exhibits a volatility of 58.9%, significantly lower than the 90.1% for an equal-weighted buy-and-hold strategy of all cryptocurrencies in the market. Furthermore, while the equal-weighted buy-and-hold portfolio shows negative skewness, exposing investors to periodic market-wide declines, the carry trade returns display positive skewness, implying a lower probability of significant downturns. This is evidenced by their ability to withstand the major market crashes in 2018 and 2021, as shown in Figure 1. This outcome is not surprising, as the combination of long and short positions in the carry trade appears to neutralize exposure to aggregate fluctuations in the cryptocurrency market.

Next, we decompose the excess returns of cryptocurrencies into two components, namely, the returns from cryptocurrency appreciation and the returns from the interest rate differential. Our results demonstrate that the significant excess returns generated by the carry trade in cryptocurrencies are caused mainly by movements in spot cryptocurrency prices, rather than interest rate spread. Additionally, our findings indicate that high-interest cryptocurrencies tend to increase in value rather than decrease relative to their low-interest counterparts, thereby refuting the validity of the uncovered interest rate parity (UIP) hypothesis within the cryptocurrency market, confirming the findings of Cong, He and Tang (2023).

Further examination shows that the carry trade returns of cryptocurrencies cannot be solely attributed to the established risk factors in the crypto markets, including the three-factor model (Liu, Tsyvinski and Wu, 2022), downside risk, volatility risk, and liquidity risk of the crypto assets. While these factors may effectively explain the returns of long-only portfolios in cryptocurrencies, they fail to account for the excess returns of the carry strategy. In fact, the cryptocurrency carry trade can even be used as a hedge against some of these factors, adding an extra layer of complexity to the puzzle. The platform risk that has long plagued crypto assets also fails to explain the disparity between high- and low-interest-rate cryptocurrencies. Given the low explanatory power of risk factors within the crypto markets, we explore if the risk factors observed in other asset

classes can achieve the desired result. The carry factor from fiat currency still fails to explain the carry trade returns.[2] Instead, we observe that the volatility risk of the equity market (i.e., VIX) plays a major role in driving the cryptocurrency carry return. In particular, high-interest cryptocurrencies experience larger declines in response to spikes in equity volatility compared to their low-interest counterparts and thus are rewarded with higher excess returns. Once the impact of equity volatility risk is considered, the model's overall explanatory power substantially increases. We find that the five-factor cryptocurrency model proposed by Cong, Karolyi, Tang and Zhao (2022) can also explain part of the carry return and that the explanatory power of equity volatility remains when controlling for the five-factor model.

We then proceed to explore how the VIX can help explain the cryptocurrency carry trade returns. Building on the work of Bekaert and Hoerova (2014), which decomposes VIX into realized volatility and variance risk premium, we explore two plausible explanations. First, the variance premium, defined as the difference between risk-neutral and objective expectations of market return variance, has long been considered a proxy for investors' risk aversion and ambiguity aversion (Rosenberg and Engle, 2002; Bakshi and Madan, 2006; Zhou, 2018; Miao, Wei and Zhou, 2019). It has been demonstrated to predict returns across equity (Bollerslev, Tauchen and Zhou, 2009), bond (Wang, Zhou and Zhou, 2013; Choi, Mueller and Vedolin, 2017), and fiat currency markets (Londono and Zhou, 2017). Thus, we test whether the variance risk premium component of the VIX could be linked to cryptocurrency carry returns. However, our findings reject this hypothesis, showing that the explanatory power of the VIX on cryptocurrency carry returns stems primarily from the realized volatility, rather than the variance risk premium component. Specifically, the high-minus-low (HML) portfolio has an insignificant loading on the variance risk premium, yet a negative loading on the realized volatility. The remaining potential explanation then lies in the realized equity volatility.

The relationship between equity returns and volatility has been well documented, as seen in Pindyck (1984) and French, Schwert and Stambaugh (1987).[3] For equity volatility to propagate to

[2]In addition, we find geopolitical risks arising from exogenous events, as suggested by Caldara and Iacoviello (2022), also fail to explain cryptocurrency carry returns.

[3]There are two primary explanations for the relationship between equity returns and volatility. French, Schwert and Stambaugh (1987) propose that the expected market risk premium is linked to stock volatility. An alternative explanation is offered by Black (1976) and Christie (1982), who suggest that leverage can induce a negative ex post relationship between returns and volatility for common stocks, even if the total firm's volatility and expected return remain constant. For instance, if a firm's volatility and expected return are constant, a decline in stock prices (relative

the cryptocurrency market, one plausible mechanism is through price contagion, as described by Capponi and Larsson (2015). Their model posits that financial institutions with target leverage ratios are forced to liquidate assets when part of their portfolio declines, potentially triggering a sell-off in other asset classes. If institutions simultaneously hold both equity and cryptocurrency carry positions, they may be compelled to unwind these carry positions to meet funding requirements when the stock market volatility increases. Under this assumption, heightened equity market volatility could prompt institutions to liquidate both equity and crypto carry positions, leading to price reactions across both equity and cryptocurrency markets.

Finally, we explore the returns generated by a time-series carry trade between stablecoins and the U.S. dollar. This popular trading strategy involves borrowing the U.S. dollar, exchanging it for stablecoins that are pegged to the U.S. dollar, and depositing them on online platforms to earn a higher interest rate. We utilize two widely used stablecoins, DAI and USDT (Tether), in our examination of the carry trade returns between stablecoins and the U.S. dollar. Our findings show that this strategy generates an annualized return of approximately 9% with a Sharpe ratio as high as 3.54. According to the preliminary evidence, it seems that the heightened returns could potentially be attributed to the depegging risks that are inherent in stablecoins.

Our paper contributes to three strands of literature. First, our study adds to the body of research on the carry trade. Aligning with the findings of Koijen, Moskowitz, Pedersen and Vrugt (2018), who expand the concept of carry trade beyond currencies and show that it generates positive returns across global equities, commodities, bonds, options, and other markets, we confirm that carry trades in cryptocurrencies also yield impressive returns. More importantly, previous studies have shown that the carry trade in fiat currency markets generates substantial positive returns (e.g., Burnside, Eichenbaum, Kleshchelski and Rebelo, 2011; Lustig, Roussanov and Verdelhan, 2011; Lettau, Maggiori and Weber, 2014). The positive carry trade return is attributed to various types of risks, such as carry trade risk (e.g., Lustig, Roussanov and Verdelhan, 2011), downside risk (e.g., Lettau, Maggiori and Weber, 2014), global foreign exchange volatility risk (e.g., Menkhoff, Sarno, Schmeling and Schrimpf, 2012), peso-event risk (e.g., Burnside, Eichenbaum, Kleshchelski and Rebelo, 2011), and jump risk (e.g., Lee and Wang, 2019), among others. However, our research

to bond prices) increases leverage, which subsequently increases both the expected return and the variance of the stock's return. In this study, we do not contribute to this debate but accept the negative relationship between equity volatility and equity returns as given.

reveals that the drivers of carry returns in fiat currencies and cryptocurrencies differ significantly, as many risk factors identified in fiat markets fail to explain cryptocurrency carry trade returns.

Second, our paper complements and extends the studies on the cryptocurrency carry trade and cryptocurrency returns more broadly. Specifically, this work differs from existing cryptocurrency carry studies in the following aspects. First, we use realistic interest rates from a major online trading platform for all cryptocurrencies. Some studies use crypto-derivatives to infer interest rates, such as perpetual swaps (Franz and Schmeling, 2021) and futures contracts (Christin, Routledge, Soska and Zetlin-Jones, 2023; Schmeling, Schrimpf and Todorov, 2023).[4] Crypto-derivatives suffer from liquidity issues (e.g., Greene and McDowall, 2018; Kumar, 2022), especially when the underlying token has a smaller market share. Moreover, Franz and Valentin (2020) show that the spot and derivatives markets for cryptocurrencies are segregated, and conventional economic parities are often violated. Second, instead of focusing on a few ad hoc tokens (e.g., Christin, Routledge, Soska and Zetlin-Jones, 2023; Schmeling, Schrimpf and Todorov, 2023),[5] our study includes over 40 cryptocurrencies. Our cryptocurrency universe consists of all actively traded currencies from the Bitfinex platform with non-trivial borrowing and lending rates, including major tokens such as Bitcoin and Ethereum, as well as stablecoins like USDT and DAI. Third, we conduct a risk factor analysis of the cryptocurrency carry trade from an asset-market perspective. For instance, Cong, He and Tang (2023) use a token-based equilibrium model to show that cryptocurrencies with higher staking reward rates attract more staking activity, leading to price appreciation compared to those with lower reward rates. Similarly, Franz and Schmeling (2021) argue that carry returns stem from capital insufficiency and highly leveraged speculation. However, we find that established risk factors from both the fiat currency and cryptocurrency markets fail to explain the observed carry trade returns. Instead, we show that changes in equity market volatility account for a significant portion of the carry returns. Our study, therefore, complements the existing literature by offering equity volatility risk as an explanation for cryptocurrency carry returns.

Our finding presents a novel perspective regarding the returns of cryptocurrencies by identify-

[4]The strategy in Christin, Routledge, Soska and Zetlin-Jones (2023) is to short the perpetual futures of Bitcoin and long Bitcoin as a hedge. Therefore, their strategy is actually an arbitrage strategy based on deviations of covered interest rate parity. However, the carry trade strategy in our paper is an arbitrage strategy based on deviations of uncovered interest rate parity.

[5]The strategy in Schmeling, Schrimpf and Todorov (2023) is to long cryptocurrencies in the spot market and short the cryptocurrency futures, which is the similar to that in Christin, Routledge, Soska and Zetlin-Jones (2023).

ing equity market volatility as a driver of cryptocurrency carry trade returns, deviating from the conventional understanding. Cryptocurrencies are often described as decentralized, less influenced by macroeconomic factors, and exhibiting low correlation with other asset classes. However, our research reveals that equity market factors affect the returns of cryptocurrencies, particularly those with high-interest-rates. This observation contributes to a better understanding of the relationship between cryptocurrencies and traditional financial markets, highlighting the importance of considering equity market factors in cryptocurrency investments.

Finally, our paper is related to the literature on return predictability of cryptocurrency. Liu, Tsyvinski and Wu (2023) find that price-to-new addresses ratios negatively predict cryptocurrency returns. Li, Liu, Liu and Zhu (2024) use machine learning methods to predict cryptocurrency return and highlight that market value to realized value ratio, new addresses, and active addresses are the most important predictors. Liu and Tsyvinski (2021) find that cryptocurrency network factors, momentum, investor attention, and investor sentiment predict cryptocurrency returns. Zhang, Li, Xiong and Wang (2021) find that downside risk predicts cryptocurrency returns. Cong, Karolyi, Tang and Zhao (2022) and Liu, Tsyvinski and Wu (2022) propose cryptocurrency factor models to explain cryptocurrency returns in the cross-section. We add to the literature by showing that carry can predict cryptocurrency returns and that this predictability at least partly comes from the exposures of cryptocurrencies to equity market volatility.

The rest of the paper is organized as follows. Section 2 describes the data. Section 3 confirms that the interest rate differential can predict cryptocurrency excess returns using panel-regression analysis. Section 4 constructs the cryptocurrency carry trade portfolios and presents some stylized facts about the distribution and dynamics of the carry trade returns. Section 5 investigates plausible economic explanations for the carry return. Section 6 analyzes the carry trade of stablecoins, and Section 7 concludes the paper.

2 Data

We obtain interest rate data for 46 cryptocurrencies (including two stablecoins) from Bitfinex, a peer-to-peer platform for lending and borrowing.[6] The sample period spans from October 2016 to

[6] We select Bitfinex as it is the largest exchange offering interest rates for Bitcoin through a limit order book, making its rates representative of the cryptocurrency lending market (Franz and Valentin, 2020).

November 2021, as data prior to October 2016 are deemed unreliable. Bitfinex facilitates margin trading, allowing investors to borrow cryptocurrencies, with the interest rates reflecting the cost of this activity.[7] Interest rates are available at hourly frequency. To mitigate the impact of outliers, we employ a precautionary measure of replacing hourly interest rates that exceed the 99th percentile of historical observations in the whole sample period with their preceding value. We then calculate the daily interest rate by taking the volume-weighted average of hourly observations, as per Franz and Valentin (2020). The final daily interest rate is utilized to calculate the monthly and weekly interest rates by multiplying it by 30 and by 7, respectively.

Investors can earn interest by depositing their cryptocurrency holdings on Bitfinex, much like in traditional fiat lending markets. Following the definition of carry for the fiat currency market (Lettau, Maggiori and Weber, 2014), we define the carry of a cryptocurrency, $carry_t$, as the interest rate differential between cryptocurrency, i_t^c, and the U.S. dollar, i_t^{USD}:

$$carry_t = i_t^c - i_t^{USD}. \tag{1}$$

We define the weekly (monthly) excess return of a cryptocurrency as the total return earned by borrowing the U.S. dollar for a week (month), using the funds to purchase cryptocurrency, and then depositing the cryptocurrency on Bitfinex for the same period. The excess return includes two components: the interest rate differential between the cryptocurrency and the U.S. dollar, and the price change of the cryptocurrency during the week (month). Thus, the excess return of an individual cryptocurrency (in log) is given by

$$r_{t+1} = carry_t + s_{t+1} - s_t \,, \tag{2}$$

where s_t is the log price of cryptocurrency in U.S. dollars at time t. This is consistent with the calculation of excess return for fiat currency in Lettau, Maggiori and Weber (2014). The price of cryptocurrency in U.S. dollars is similar to the exchange rate in the fiat currency market. The detailed discussion on the interest rate, excess return, and cryptocurrency carry can be found in Appendixes A and B.

[7] The deposit rate is set at 85% of the borrowing rate, with the platform taking a 15% fee on earnings. For example, if a borrower pays 10% interest, the fund provider earns 8.5% after fees.

The summary statistics of the annualized interest rates of borrowing cryptocurrencies are illustrated in Panel A of Table 1. Bitcoin, Ethereum, Litecoin, and USDT (Tether) are among the most popular cryptocurrencies. The means of their interest rates are 8.26%, 6.04%, 6.22%, and 12.03%, respectively. The standard deviations of their interest rates are 0.50%, 0.29%, 0.50%, and 0.47%, respectively.

[Insert Table 1 Here]

We collect daily spot prices, market capitalization, and trading volume of cryptocurrencies from coinmarketcap.com as per Liu, Tsyvinski and Wu (2022). To ensure accuracy, we eliminate the first-week observations following the launch of each cryptocurrency as suggested by Cong, He and Tang (2023), who posit that abnormal fluctuations in cryptocurrency prices occur during this period. We winsorize spot prices by replacing their observations larger than the 99th percentile or smaller than the 1st percentile with the previous values. Panel B of Table 1 presents a summary of daily log price changes of cryptocurrencies. Except for stablecoins, the prices of cryptocurrencies are quite volatile. The daily standard deviations are around 7% for most cryptocurrencies. In addition, most cryptocurrencies appreciate during the sample periods. For example, Bitcoin, Ethereum, and Litecoin appreciate 0.24%, 0.31%, and 0.21% per day on average, respectively.

We acquire the monthly interest rate of the U.S. dollar from Federal Reserve Economic Data (FRED). We source the three factors of weekly market, size, and momentum factors of cryptocurrencies from Liu, Tsyvinski and Wu (2022), five factors of weekly market, size, momentum, value, and network factors of cryptocurrencies from Cong, Karolyi, Tang and Zhao (2022); daily geopolitical risk data from Caldara and Iacoviello (2022); daily VIX and realized volatility of the U.S. stock market (S&P 500 Index) from CBOE; and daily fiat currency spot and forward exchange rates from DataStream.

3 Predictive Power of Carry

In this section, we analyze whether the carry can predict future excess returns in the cryptocurrency market. This approach follows the predictive regression framework introduced by Fama (1984) and further developed by Koijen et al. (2018). Specifically, we examine the following predictive

regression for cryptocurrencies:

$$r_{t+1} = \alpha + \beta(i_t^c - i_t^{USD}) + \epsilon_{t+1}\,, \tag{3}$$

where r_{t+1} denotes the log excess return of a cryptocurrency from time t to time $t+1$, as defined in Equation (2). The terms i_t^c and i_t^{USD} represent the interest rates for the cryptocurrency and the U.S. dollar, respectively, over the same time period. In other words, the right-hand side of this regression represents the carry.

A conventional way to interpret this regression is as a test of uncovered interest rate parity (UIP), as proposed by Fama (1984). The UIP hypothesis, originally introduced by Keynes (1923), suggests that a currency with a higher interest rate should depreciate relative to a currency with a lower interest rate. Therefore, in a UIP world, investors cannot profit from borrowing in a low-interest-rate currency and lending in a high-interest-rate currency. Fama (1984) posited that under UIP, the coefficient α in Equation (3) should be zero, and the estimate for β should also be zero, implying that the interest rate differential $i_t^c - i_t^{USD}$ cannot predict the future excess returns of cryptocurrencies. Consequently, our test can be viewed as an empirical investigation of UIP's validity in the cryptocurrency market.

While this is an important interpretation, one might intuitively argue that cryptocurrencies are unlikely to conform to the UIP. This is because an essential implication of UIP is that currencies have zero excess returns, i.e., no risk premium is associated with holding a currency.[8] While traditional fiat currencies, such as the British pound, may exhibit no significant risk premium relative to the U.S. dollar, it is less plausible to suggest that cryptocurrencies, which are inherently more volatile, command no risk premium for investors.

In this context, the primary focus of our analysis is to test whether the carry (i.e., the interest rate differential between cryptocurrencies and the U.S. dollar) can predict the log excess returns of cryptocurrencies, in line with the methodology of Koijen et al. (2018). Furthermore, we investigate the relationship between carry, expected price changes, and total expected returns in greater detail. Specifically, as hypothesized by Koijen et al. (2018), if $\beta = 0$, carry does not predict returns, which aligns with a generalized form of UIP. On the other hand, $\beta = 1$ implies that expected returns

[8] One can verify this by setting $\alpha = \beta = 0$ in Equation (3).

move one-for-one with carry. While $\beta = 0$ indicates that total returns are unpredictable, $\beta = 1$ suggests that price changes (returns excluding carry) are unpredictable based on carry. If $\beta < 0$, this would indicate that carry predicts a negative price change that more than offsets the direct effect of a positive carry. If $0 < \beta < 1$, this implies that a positive carry is associated with a negative expected price appreciation, with the market partially reversing the carry, though not entirely. Finally, $\beta > 1$ would suggest that a positive carry is associated with positive expected price appreciation, allowing investors to benefit from both carry and price appreciation, i.e., carry predicts further price increases.

To empirically test these hypotheses, we conduct a panel regression analysis using data from 44 cryptocurrencies.[9] We examine the relationship between interest rate differentials and excess returns using both weekly and monthly return frequencies. To maximize the number of observations, we conduct the analysis at a daily frequency. Specifically, each day, we simulate a strategy where U.S. dollars are borrowed, converted into cryptocurrency, deposited on the Bitfinex platform for one month (or one week), and then converted back into U.S. dollars, with the loan being repaid. This process is repeated daily, allowing us to calculate monthly (or weekly) excess returns on a daily basis.

Table 2 presents the results of the panel regression. The estimates for β are significantly positive for both weekly and monthly returns, even after controlling for individual and time fixed effects. Notably, in the model incorporating both individual and time fixed effects, a 1% increase in the interest rate differential is associated with a 1.69% (1.66%) increase in the future excess return of the cryptocurrency. The fact that $\beta > 1$ indicates that carry predicts further price increases, reinforcing the notion that the investors holding high-interest cryptocurrencies can benefit from both high deposit rates and future price appreciation. These findings suggest that, similar to fiat currencies, the interest rate differential between cryptocurrencies and the U.S. dollar can positively predict future excess returns in the cryptocurrency market.

[Insert Table 2 Here]

[9] We exclude the two stablecoins, DAI and USDT, from our analysis because their prices are pegged to the U.S. dollar, and thus their carries are not expected to predict their excess returns.

4 Performance of Cryptocurrency Carry Trade

In this section, we examine the application of carry trade strategies to cryptocurrencies. A carry trade strategy consists of borrowing low-interest-rate currencies and investing in high- interest-rate currencies. Section 4.1 explains how we construct the cryptocurrency carry trade portfolios. Section 4.2 analyzes the historical performance of these portfolios.

4.1 Construction of Cryptocurrency Carry Trade Strategies

We investigate the cross-sectional carry trade within a universe of 44 cryptocurrencies, excluding stablecoins, as their values are relatively constant.[10] Similar to the carry trade in fiat currencies, this strategy goes long on cryptocurrencies with high-interest-rates and short on cryptocurrencies with low-interest-rates. At the end of each week, we rank the cryptocurrencies by their interest rates i and sort them into three portfolios (Low, Mid, and High portfolios), with the Low portfolio containing tokens with the lowest interest rates and the High portfolio containing tokens with the highest interest rates. To make sure that our results are not driven by major cryptocurrencies, we allocate an equal weight of cryptocurrencies to each portfolio. We hold the portfolios for one week and rebalance at the end of the week.[11] The carry trade return is defined as the difference in returns between the High portfolio and the Low portfolio.

Table 3 lists the top five most frequently occurring cryptocurrencies in the High and Low portfolios, along with the start and end dates of the sample period for each cryptocurrency; the total number of weeks in the sample; the number of weeks in Low, Mid, and High portfolios; and their market capitalization. Bitcoin Gold, Bitcoin, Neo, Ethereum Classic, and Ethereum appear the most often in the High portfolio as they generally have relatively high-interest-rates. In contrast, IOTA, Santiment Network Token, Unus Sed LEO, Zcash, and EOS are the most frequent in the Low portfolio. Notably, most cryptocurrencies switch between portfolios over time, with the exception of Unus Sed LEO, which remains consistently in the Low portfolio throughout the

[10] In Section 6, we analyze the carry trade returns separately using stablecoins.

[11] Following Liu, Tsyvinski and Wu (2022), we divide the year into 52 weeks. The first week consists of the first seven days, and each of the first 51 weeks consists of seven days. The last week includes the remaining days. Since interest rates of the U.S. dollar and VIX are not available on weekends, we rebalance the portfolio on the last working day of each week. Although the last date in our sample, November 30, 2021, is not the last working day of the week, we use it to calculate the return of cryptocurrencies for the last week of our sample.

sample period.

[Insert Table 3 Here]

4.2 Stylized Facts of Cryptocurrency Carry Trade Returns

Table 4 reports the historical performance of the cryptocurrency carry trade. The first three columns correspond to the cryptocurrency portfolios with low, mid, and high-interest-rates, respectively. The fourth column shows the return of the high-minus-low (HmL) portfolio. We also report the summary statistics of an equal-weighted buy-and-hold cryptocurrency portfolio in the last column. The average returns, standard deviations, and Sharpe ratio are annualized quantities and expressed in percentage points.

The last column reveals that the excess returns of the equal-weight buy-and-hold cryptocurrency portfolio are large yet highly volatile. The average annualized excess return across all cryptocurrencies is over 60%. However, this attractive return is accompanied by high volatility, with an average annualized volatility of approximately 90%. This confirms the speculative nature of cryptocurrency and the sizable risk premium associated with it.

The average excess returns of cryptocurrencies across carry portfolios exhibit a monotonically increasing pattern with interest rates. Specifically, high-interest-rate cryptocurrencies deliver an average excess return of 93.9%, compared to 50.4% for their low-interest-rate counterparts. Despite the difference in returns, the volatilities of the high and low portfolio are similar, with high-interest-rate cryptocurrencies exhibiting volatility of 102.7% and low-interest-rate ones at 93.8%. The cryptocurrency carry trade strategy, which takes short positions in the low-interest-rate cryptocurrencies and long positions in the high-interest-rate ones, has delivered a decent performance, generating an annualized return of 43.4%. Notably, the carry trade strategy exhibits lower volatility than the EW strategy, which buys and holds all cryptocurrencies, with an annualized volatility of 58.9% versus 90.1% for the latter. By constructing a long-short portfolio in cryptocurrencies, the carry trade substantially reduces the exposure to overall cryptocurrency market fluctuations, resulting in lower volatility. Nevertheless, the returns for the cryptocurrency carry trade are significantly larger and more volatile than those for the fiat currency carry trade, which has an average return of 5.29% and a volatility of 7.80% according to Koijen, Moskowitz, Pedersen and Vrugt

(2018).

We decompose the cryptocurrency excess return into two components: the return from the interest rate differential between the cryptocurrency and the U.S. dollar, known as the carry, and the return from currency appreciation relative to the U.S. dollar (Δs). As expected, return on interest rate differential increases from the Low to High portfolios, with the high-interest-rate portfolio benefiting from positive carry due to its higher-than-USD interest rates, while the low-interest-rate portfolio experiences negative carry from lower-than-USD interest rates. Interestingly, the return from price appreciation also follows an upward-sloping trend across carry portfolios. As the excess return for each portfolio is the sum of its carry and price appreciation, the high-interest-rate portfolio generates the highest excess return, driven by both higher carry and greater price appreciation.

The results provide several insights into crypto assets. First, changes in spot cryptocurrency prices contribute more to carry trade returns than interest rate differentials. Specifically, out of the 43.4% of total excess return, 24% of it can be attributed to appreciation in cryptocurrency values, and the remaining 19.4% comes from the interest rate differentials. Second, the uncovered interest parity (UIP) hypothesis has been observed to be inapplicable in the cryptocurrency market. According to UIP, a positive interest differential should be offset by the future depreciation of currencies. However, the cryptocurrency market shows that high-interest cryptocurrencies tend to appreciate rather than depreciate relative to their low-interest counterparts, further widening the return gap. Finally, carry is a strong predictor of excess returns in the cryptocurrency market. A higher carry is associated with a higher excess return, indicating that carry is a useful tool for predicting the performance of cryptocurrencies.

Moreover, the Sharpe ratio of the carry return exceeds that of the equal-weighted portfolio return, being 0.74 compared to 0.70, respectively. This supports the findings documented in Koijen, Moskowitz, Pedersen and Vrugt (2018) that the carry portfolio, in most cases, has a higher Sharpe ratio. In addition, the returns on an equal-weighted buy-and-hold cryptocurrency portfolio exhibit negative skewness (-0.09), suggesting that investors are exposed to a heightened risk of aggregate cryptocurrency market crashes. In contrast, the carry return is positively skewed. The cryptocurrency carry strategy exhibits excess kurtosis, substantially larger than the kurtosis of the equal-weighted portfolio, indicating fat-tailed positive and negative returns. It is interesting to

observe positive skewness and large kurtosis at the same time. As illustrated in Koijen, Moskowitz, Pedersen and Vrugt (2018), a typical example is that the strategy suffers extreme negative returns, particularly around adverse crypto-events, which are then followed by even more extreme positive returns during the recovery (resulting in overall positive skewness). In short, the higher moments of the cryptocurrency carry strategy imply the presence of extreme returns, with larger positive returns than negative ones. This dynamic highlights the strategy's speculative nature, with the potential for significant gains following large losses.

[Insert Table 4 Here]

To visualize this observation, Figure 1 depicts the cumulative returns of the carry portfolio and the equal-weighted (EW) portfolio. For ease of comparison, we scale the carry portfolio return to match the ex post volatility (90.1% annualized) of the equal-weighted portfolio return, as per Koijen, Moskowitz, Pedersen and Vrugt (2018).[12] Over the sample period, the carry return demonstrates a steady upward trend, albeit with a few slight dips around 2018 and the start of 2021. On the other hand, the EW return is characterized by more pronounced fluctuations, particularly around 2018 and 2021. Of particular interest is the behavior of the two returns during cryptocurrency market crashes. The carry return experiences a moderate decline prior to the 2018 crash, but it manages to withstand the impact of the bear market. On the contrary, the EW return exhibits a sharp surge before the 2018 crash and suffers a significant decline during the crash. Similar patterns are observed before and during the 2021 crash. The findings are in line with the notion that the carry trade entails both long and short positions, rendering it insensitive to fluctuations in the overall cryptocurrency market. This implies that it is relatively less vulnerable to crash risk in the realm of crypto markets.

[Insert Figure 1 Here]

It is tempting to assume that high-interest cryptocurrencies may offer high returns due to their inherent higher risks. These risks could arise from factors such as being smaller, being less established, and being less liquid digital assets, which would require higher compensation. In order to further investigate this claim, we conduct a detailed analysis of carry portfolios and report their characteristics in Table 4. Specifically, we study the average age, market capitalization, and trading

[12] Specifically, the scaled carry portfolio return is calculated as the raw return divided by a scaling factor. The scaling factor is the ratio of the volatility of the carry portfolio return to the volatility of the equal-weighted portfolio return.

volumes of the three cryptocurrency portfolios.

Our findings, as shown in Table 4, indicate that the performance of carry returns is not driven by these characteristics. While high-interest cryptocurrencies tend to be slightly younger (average age of 74.9 weeks) than low-interest cryptocurrencies (77.5 weeks), they do not appear to be smaller or illiquid. There is no clear pattern in age, market capitalization, and trading volumes across carry portfolios, suggesting that these characteristics are not the primary drivers of the carry trade returns. In the subsequent sections, we aim to determine whether the cross-sectional carry trade returns can be explained by other established risk factors.

5 Carry Trade Returns and Risk Factors

In this section, we examine different risk factors to explain cryptocurrency carry return. We start by examining common risk factors in the cryptocurrency market, and then we study risk factors in the fiat currency market and equity market.

5.1 Risk Factors of Cryptocurrency Market

5.1.1 Cryptocurrency Common Factors

The study by Liu, Tsyvinski and Wu (2022) identifies three common risk factors for cryptocurrencies: market, size, and momentum. The market factor is the weighted average return of all cryptocurrencies in the market minus the risk-free interest rate, which captures the systematic market risk. The size factor is the return of a portfolio that longs small cryptocurrencies and shorts large cryptocurrencies.[13] The momentum factor is the return of a portfolio that longs cryptocurrencies with large increases in prices and shorts those with small increases in prices. They demonstrate that these factors can effectively explain the abnormal returns of various cryptocurrency trading strategies. Do different expected returns from the carry portfolios represent different exposure to these common cryptocurrency factors? To answer this question, we estimate the following time-series

[13] Liu, Tsyvinski and Wu (2022) use market capitalization, price at the end of the week, and highest price of the week to proxy for size.

regression:

$$r_{i,t} = \alpha + \beta_1 f_{cmkt,t} + \beta_2 f_{csize,t} + \beta_3 f_{cmom,t} + \varepsilon_{i,t}, \quad i = \text{Low}, \text{Mid}, \text{High}, \text{HmL} \tag{4}$$

where $r_{i,t}$ is the excess return of portfolio i in week t.[14]

Panel A of Table 5 presents the regression results. Betas on the cryptocurrency market factor are significant across the Low, Mid, and High carry portfolios, with coefficients hovering around 0.8. Interestingly, the Low and Mid portfolios exhibit higher market beta than the High portfolio. This unexpected trend contradicts the anticipated pattern if carry trade returns were a risk premium of market risk, in which case betas would increase progressively from the Low to the High portfolio. Consequently, the high-minus-low portfolio displays a negative exposure to the market factor, although this observation lacks statistical significance. Similar to the market beta, the betas on the cryptocurrency size factor display a decreasing trend from the Low to the High portfolio, making the high-minus-low portfolio a hedge against the crypto-size factor. While the high-minus-low carry return hedges against the market and size factor, it has a positive exposure to the momentum factor. However, the betas on the momentum factors display a V-shape trend from the Low to the High portfolio, and only the beta of the Mid portfolio is statistically significant at the 10% level.

The risk-adjusted return after incorporating the three cryptocurrency factors is 0.9%, which remains statistically significant and an even higher figure than the raw return of 0.8% (43.4% annualized return as reported in Table 4). Despite the cross-section portfolios having an R^2 of approximately 0.6, indicative of the common nature of these factors in cryptocurrency assets, the high-minus-low portfolio's R^2 is as low as 0.045. It suggests that while the common factors can explain long-only portfolios of cryptocurrencies, they fail to account for the abnormal returns of the carry strategy, potentially adding to its puzzling nature.

In addition to Liu, Tsyvinski and Wu (2022)'s three factors, Cong, Karolyi, Tang and Zhao (2022) provide five common factors in the cryptocurrency market: market, size, momentum, value, and network.[15] We test whether these factors can explain the cryptocurrency carry return by

[14]The log excess return of portfolio i is the equal-weighted average of the log excess returns of cryptocurrencies in portfolio i. The log excess return of an individual cryptocurrency is defined in Equation (2).

[15]The market factor represents the return of the cryptocurrency market index minus the risk-free interest rate. The size factor captures the return of a portfolio that longs cryptocurrencies with the smallest market capitalization and

estimating the following time-series regression:

$$r_{i,t} = \alpha + \beta_1 f_{Mkt,t} + \beta_2 f_{Size,t} + \beta_3 f_{Momentum,t} + \beta_4 f_{Value,t} + \beta_5 f_{Network,t} + \varepsilon_t, \quad (5)$$
$$i = \text{Low}, \text{Mid}, \text{High}, \text{HmL}$$

where $r_{i,t}$ is the excess return of portfolio i in week t.

Panel B of Table 5 presents the results. Compared to the three-factor model, the five-factor model does not seem to capture more common variations of Low, Mid, and High portfolio returns, as the adjusted R^2 is lower, approximately 0.4, compared to those in Panel A. Similar to the results in Panel A, the coefficients of the Low, Mid, and High portfolios for the market factor are significantly positive, while the size and momentum factors are statistically insignificant. The coefficients of the HML portfolio returns on the market, size, and momentum remain insignificant, indicating that exposures to these three factors are unlikely to drive the cryptocurrency carry return.

The inclusion of the additional two factors—the value factor and network factor—shows a monotonic increase in their coefficients across Low, Mid, and High portfolio returns, although many of these coefficients are statistically insignificant. Interestingly, the coefficients of the HML portfolio return on these two factors are significantly positive. The abnormal return, measured by the constant term in the regression, becomes insignificant after incorporating these two new factors.

[Insert Table 5 Here]

These findings suggest that the five-factor model provides some explanatory power for cryptocurrency carry returns. Specifically, cryptocurrencies with higher interest rates tend to have greater exposure to the value and network factors, although the statistical significance remains inconclusive.

5.1.2 Cryptocurrency Volatility and Downside Factor

Following the literature on fiat currency carry trade (Menkhoff, Sarno, Schmeling and Schrimpf, 2012; Lettau, Maggiori and Weber, 2014), we examine whether the risk factors that capture higher

shorts those with the highest. The momentum factor is based on the return of a portfolio that longs cryptocurrencies with the highest returns over the past two weeks and shorts those with the lowest. The value factor reflects the return of a portfolio that longs cryptocurrencies with the lowest returns over the past 52 weeks and shorts those with the highest. Lastly, the network factor represents the return of a portfolio that longs cryptocurrencies with the highest growth rate in total addresses with balances and shorts those with the lowest growth rate.

moments of the cryptocurrency market, namely the volatility risk and downside risk, can explain carry trade returns in the cryptocurrency market.

We construct a weekly cryptocurrency volatility risk factor following Menkhoff, Sarno, Schmeling and Schrimpf (2012),

$$\sigma_t^{crypto} = \frac{1}{T_t} \sum_{d \in T_t} \Big(\sum_{j \in N_d} \frac{|\Delta s_{j,d}|}{N_d} \Big), \tag{6}$$

where $\Delta s_{j,d}$ is the change in log price of cryptocurrency j on day d. N_d is the total number of cryptocurrencies on day d. T_t is the total number of trading days in week t. σ_t^{crypto} is the cryptocurrency counterpart of the FX volatility factor as proposed in Menkhoff, Sarno, Schmeling and Schrimpf (2012). It measures the realized volatility in week t of the cryptocurrency market. We define the innovations of σ_t^{crypto} as

$$\Delta\sigma_t^{crypto} = \sigma_{t+1}^{crypto} - \sigma_t^{crypto}. \tag{7}$$

Following the method of Henriksson and Merton (1981), we construct a cryptocurrency downside risk factor as

$$f_{cds,t} = \max(0, -r_{m,t}), \tag{8}$$

where $r_{m,t}$ is the cryptocurrency market return at time t. Notice that f_{ds} only takes nontrivial values when the cryptocurrency market has realized a loss and, therefore, captures the downside of the cryptocurrency market.

To test whether the downside risk and volatility risk can explain the cryptocurrency carry trade, we first run the following univariate regressions for all carry portfolios:

$$p_{i,t} = \beta_0 + \beta_1 f_t + \varepsilon_t, \quad i = \text{Low, Mid, High, HmL}, \quad f_t = f_{cds,t},\ \Delta\sigma_t^{crypto}, \tag{9}$$

and then run multivariate regressions with the common factors as control variables:

$$p_{i,t} = \beta_0 + \beta_1 f_t + \beta_x X_t + \varepsilon_t, \quad i = \text{Low, Mid, High, HmL}, \quad f_t = f_{cds,t},\ \Delta\sigma_t^{crypto} \tag{10}$$

where X denotes a vector of control variables $X_t = (f_{cmkt,t}, f_{csize,t}, f_{cmom,t})$.

Note that both the volatility and downside risk are countercyclical factors by construction,

implying that higher values indicate a more unfavorable scenario. Therefore, positive loadings on these factors suggest that the asset earns higher returns when the risk is elevated, thus serving as a hedge against that risk.

Panels C and D of Table 5 present the results with the cryptocurrency volatility factor and the cryptocurrency downside risk factor, respectively. The univariate regressions show that all portfolios exhibit negative loadings on both volatility and downside risk, suggesting that these portfolios generally experience lower returns when cryptocurrency market volatility risk or downside risk is high. However, the high-minus-low portfolio demonstrates a positive but insignificant exposure to the volatility factor. This result implies that instead of compensating investors for taking volatility risk, the carry strategy acts as a hedge against this type of risk. Additionally, the exposure of the high-minus-low portfolio to the downside risk factor is negative, suggesting that the carry strategy suffers losses when downside risk increases. However, the coefficient is statistically insignificant.

Despite the suggestive pattern, the betas do not show a monotonic trend from the Low to High portfolios in both univariate and controlled regressions. The betas even change signs in certain cases when controlling for cryptocurrency common factors. We therefore conclude that the returns from cryptocurrency carry trade do not represent a risk premium for either volatility or downside risk.

5.1.3 Platform Risk

The literature reveals the instability of cryptocurrency platforms. For example, Sockin and Xiong (2023) show that the network effect can cause the cryptocurrency market to melt down and highlight the fragility of cryptocurrency. It is argued that returns on cryptocurrencies may be attributed to platform collapse risk (e.g., Ma, Tu and Zhu, 2023). Therefore, the positive carry return of cryptocurrency might also be caused by this type of risk. According to this hypothesis, if carry portfolio returns are indeed influenced by platform risk, the High portfolio should contain tokens that are more vulnerable to such risks.

Platform collapse, similar to a peso problem, is challenging to pin down due to its infrequent occurrences in historical samples. In other asset markets, the preferred method for assessing this type of risk usually involves options and other forward-looking derivatives, which incorporate event

risks into their pricing, even if those risks may not necessarily materialize. However, the absence of actively traded options for every cryptocurrency makes it even more challenging to gauge event risk for cryptocurrencies.

To test whether carry returns represent compensations for platform risks, we seek anecdotal evidence of platform collapses within our sample. If a cryptocurrency is susceptible to platform risk, we expect to see a large drop in value shortly after a platform collapse incident. To measure the susceptibility to platform risk for each cryptocurrency, we employ an event study methodology.

Define the cumulative abnormal return (CAR) of portfolio i associated with a platform collapse event that takes place at t^* as $\sum_{t=t^*-j}^{t^*+j} [p_{i,t} - E(p_{i,t})]$, where j is the window size. Here, $E(p_{i,t})$ is the expected return and is calculated using $E(p_{i,t}) = \hat{\alpha}_i + \hat{\beta}_i f_{cmkt,t}$, where $\hat{\alpha}$ and $\hat{\beta}$ are estimated using OLS regressions with data 30 weeks prior to the event.

We identify six events associated with platform collapse, and calculate the average CAR for each carry portfolio for every event.[16] We consider one week, two weeks, and four weeks before and after each event as the event window, respectively. The results are shown in Panel E of Table 5. Indeed, cryptocurrencies are subject to platform risk. Almost all portfolios have negative CARs during platform collapse events. However, the CARs of the Low to High portfolio exhibit a V-shape for the one-week and two-week windows and an almost flat pattern for the four-week window. Moreover, the CAR of the high-minus-low portfolio lacks statistical significance. Based on these results, we conclude that while cryptocurrencies are susceptible to platform collapse risk, the evidence that carry trade returns are compensations for platform risk is insufficient.

5.1.4 Liquidity Risk

To address the concern that cryptocurrency carry returns may be driven by liquidity risk, we follow the method of Brauneis, Mestel, Riordan and Theissen (2021) to construct an illiquidity risk factor for the cryptocurrency market, using the illiquidity measure proposed by Amihud (2002). We then

[16] The events we identified are associated with shutting down or suspension of cryptocurrency trading platforms due to reasons including hacking, regulation, allegations, etc. The events include: (1) BTC-e was shut down on July 25, 2017 due to allegations of illegal activities. (2) On January 16, 2018, Bitconnect was shut down due to accusations of operating as a Ponzi scheme. (3) On January 26, 2018, Coincheck was hacked and temporarily suspended several services. (4) On January 28, 2019, the Quadriga was shut down due to liquidity issues. (5) On February 5, 2019, the Quadriga applied for bankruptcy protection. (6) On May 15, 2019, Cryptopia was shut down following a hack.

examine whether this factor can explain the cryptocurrency carry returns.

$$Illiquid_t = \frac{1}{N}\Sigma_{i=1}^{N}\frac{|\Delta s_{i,t}|}{volume_{i,t}}\,, \tag{11}$$

where $\Delta s_{i,t}$ and $volume_{i,t}$ represent the change in price and the trading volume of token i on day t, respectively. Trading volumes of cryptocurrencies are measured in thousands of U.S. dollars. N denotes the number of tokens on day t. We calculate the weekly illiquidity factor as the weekly average of the daily illiquidity factor.

We add the illiquidity factor to the regression and report the results in Panel F of Table 5. The univariate regressions show that the coefficients of illiquidity factor for Low, Mid, High, and HML portfolios are all insignificant, with R^2 values closer to zero. After controlling for the three cryptocurrency factors of Liu, Tsyvinski and Wu (2022), the coefficients remain insignificant. Therefore, we conclude that cryptocurrency carry returns are not driven by liquidity risk in the cryptocurrency market.

5.2 Fiat Currency Carry Factor

Given that both the cryptocurrency carry trade and the fiat carry trade involve a carry trade strategy, it is reasonable to anticipate that the cryptocurrency carry trade may exhibit similarities with its fiat currency counterpart. In this section, we investigate whether the fiat currency carry trade can explain the anomalies observed in the cryptocurrency carry trade.

We follow Lustig, Roussanov and Verdelhan (2011) to construct the fiat currency carry trade using spot foreign exchange rates and currency forward rates. Specifically, we sort fiat currencies into six portfolios by forward discount ($s_t - f_t^w$), with s_t as the log spot exchange rate and f_t^w as the log forward rate with a one-week expiration.[17] To be consistent with the cryptocurrency carry trade, we also rebalance the fiat carry trade on a weekly basis. The fiat currency carry factor is defined as the return of the portfolio with the highest interest rates minus that with the lowest interest rates.

Univariate and multivariate regression with the fiat carry factor is presented in Panel G of Table

[17]Since we only have data of prices for futures which expire in one month (f_t^m), we estimate f_t^w with the formula $f_t^w = (f_t^m - s_t) * \frac{N}{30} + s_t$, where N is the number of days until the next rebalance day. Most of the time N =7, but it will change in value based on the rebalance dates as described in Section 4.1.

5. We observe a V-shape pattern of betas on the fiat carry factor across the Low to High portfolios, and the exposure for the high-minus-low portfolio is positive. However, the coefficients are too noisy to establish any statistical significance. Notably, when we introduce controls for the common factors in cryptocurrency markets, the loadings on the fiat carry factor change dramatically for all portfolios. Consequently, we conclude that although there seems to be a potential connection between the cryptocurrency carry trade and the fiat carry trade, the relationship is too weak to be reliably identified.

5.3 Equity Market Volatility Risk

5.3.1 VIX

Evidence suggests that cryptocurrencies display a decentralized nature, with a noticeable trend of investors shifting their portfolios toward cryptocurrencies during periods of macroeconomic and monetary policy uncertainties (e.g., Urquhart and Zhang, 2019; Rubbaniy, Khalid and Samitas, 2021). The flight-to-crypto behavior of investors may cause turbulence in traditional markets to be priced in the cryptocurrency market. In exploring the potential relationship between the cryptocurrency carry trade and equity market factors, we employ the CBOE VIX index as a metric for assessing equity market turbulence. The CBOE VIX index reflects option-implied forward-looking volatility of the S&P 500 index, typically elevated during periods of low equity returns. Let ΔVIX represent the first difference of the VIX index: $\Delta VIX_{t+1} = VIX_{t+1} - VIX_t$. Similar to our previous analysis, we conduct univariate and controlled regression analysis on ΔVIX and present the results in Table 6.

Table 6 highlights the significance of the VIX as a driving factor for carry portfolio returns. Panel A shows the results of univariate regressions and multivariate regressions after controlling for cryptocurrency three factors from Liu, Tsyvinski and Wu (2022). Panel B shows the results of multivariate regressions with controls of cryptocurrency five factors from Cong, Karolyi, Tang and Zhao (2022). In the univariate regressions, all three carry portfolios have highly significant negative exposure to VIX risk. Moreover, the coefficients on the VIX factor across the Low to High portfolios follow a monotonically decreasing pattern, suggesting that cryptocurrencies with higher interest rates earn lower returns when the VIX is high. Notice that the Low portfolio also has a

negative exposure to VIX, contrary to the popular flight-to-crypto belief, which suggests a positive relationship. This indicates that these tokens are not necessarily safe-haven assets in the sense of having good performance during equity market turmoil. Rather, they are only less susceptible to equity market downturns compared to their high-interest-rate counterparts.

When controlling for the three cryptocurrency factors of Liu, Tsyvinski and Wu (2022), the coefficients of VIX across the Low, Mid, and High portfolios become less pronounced, but the pattern remains similar, with the coefficient of the High portfolio being the lowest. Similarly, when controlling for the five cryptocurrency factors of Cong, Karolyi, Tang and Zhao (2022), the downward-sloping pattern of VIX betas persists, although the betas of the Low, Mid, and High portfolios become less significant compared to the univariate regressions. In both the univariate and controlled regressions, the high-minus-low portfolio exhibits a significantly negative beta on the VIX factor. Including the VIX factor in the regression for the high-minus-low portfolio increases the R^2 from 4.5% in the baseline regression with the cryptocurrency three factors to 5.6%, and from 10.7% in the baseline regression including the cryptocurrency five factors to 11.3%, respectively. Therefore, we conclude that a portion of the cryptocurrency carry returns can be regarded as compensation for equity market volatility risk.

[Insert Table 6 here]

5.3.2 Decomposition

To further investigate the role of the VIX in explaining cryptocurrency carry returns, we follow the methodology of Bekaert and Hoerova (2014) and decompose the VIX into realized volatility and variance risk premium to assess their explanatory power separately.

The variance risk premium, defined as the difference between the risk-neutral and objective expectations of market return variance, has long been recognized as a proxy for the risk aversion or ambiguity aversion of the representative agent (Rosenberg and Engle, 2002; Bakshi and Madan, 2006; Zhou, 2018; Miao, Wei and Zhou, 2019). Previous studies have established its relevance to returns in equity (Bollerslev, Tauchen and Zhou, 2009), fixed income (Wang, Zhou and Zhou, 2013; Choi, Mueller and Vedolin, 2017), and fiat currency markets (Londono and Zhou, 2017). Given this, it is worthwhile to examine whether the variance risk premium component of the VIX is linked

to cryptocurrency carry returns. If the variance risk premium drives cryptocurrency carry trade returns, we would expect cryptocurrencies with higher interest rates to exhibit greater sensitivity to fluctuations in the variance risk premium, while those with lower interest rates would show less exposure to such movements. Otherwise, the explanatory power of the VIX on cryptocurrency carry returns would come from realized volatility, rather than the variance risk premium component.

To quantify the variance risk premium, we compute it using the following equation:

$$VRP_t = VIX_t^2 - RVOL_t^2, \tag{12}$$

where VRP_t and $RVOL_t$ represent the variance risk premium and realized volatility on day t, respectively. According to Bollerslev, Tauchen and Zhou (2009), realized volatility is calculated as the standard deviation of the S&P 500 index during the past one month.

We present the test results in Table 7. Panels A and B report the regression results for realized volatility,[18] controlling for the cryptocurrency three-factor model from Liu, Tsyvinski and Wu (2022) and the cryptocurrency five-factor model from Cong, Karolyi, Tang and Zhao (2022), respectively. Panels C and D provide the corresponding results for the variance risk premium. Our analysis reveals that all portfolios—Low, Mid, and High—exhibit negative loadings on realized volatility, with the High portfolio demonstrating the smallest loading. Consequently, the high-minus-low portfolio shows a significantly negative coefficient on realized volatility. In contrast, while the variance risk premium occasionally displays some explanatory power, the results are inconsistent. Specifically, when controlling for the five cryptocurrency factors, the high-minus-low portfolio's coefficient for the variance risk premium is insignificant, whereas the coefficient for realized volatility remains significantly negative across both the three- and five-factor models. This evidence suggests that the explanatory power of the VIX is primarily derived from realized volatility rather than the variance risk premium. To assess the robustness of these findings, we include both realized volatility and the variance risk premium in the regression. As shown in Panel E and Panel F, the previous results remain intact.

[Insert Table 7 Here]

[18] To address the issue of non-stationary of realized volatility, we use weekly changes of realized volatility in the regression.

5.3.3 Discussion

The challenge, then, is to understand how realized equity volatility could be linked to cryptocurrency carry returns. Although no theoretical model has yet been proposed to directly link equity volatility with cryptocurrency carry returns, the relationship between realized volatility and equity returns is well documented, as noted by Pindyck (1984) and French, Schwert and Stambaugh (1987). Previous literature shows that increases in equity volatility typically coincide with declines in equity prices through the leverage channel proposed by Black (1976) and Christie (1982). This channel suggests that a decline in stock prices (relative to bond prices) increases firm leverage, which in turn raises both the expected return and the variance of the stock's return. While this explanation is intuitive for equities, it is not directly applicable to cryptocurrencies because, unlike firms, leverage is undefined for cryptocurrencies. Another possible explanation is the volatility feedback effect, as discussed by Bekaert and Wu (2000) and Wu (2001). This theory posits that heightened volatility makes investors reluctant to hold equity assets, prompting them to demand a higher risk premium. In our context, for the volatility feedback effect to affect the cryptocurrency market, one plausible mechanism is price contagion, as described by Capponi and Larsson (2015). Their model suggests that financial institutions with target leverage ratios may be forced to liquidate assets when a portion of their portfolio declines, potentially triggering sell-offs in other asset classes. If institutions hold both equity and cryptocurrency carry positions, they may be compelled to unwind these positions to meet funding requirements when stock market volatility rises. Under this assumption, increased equity market volatility could lead to the liquidation of both equity and cryptocurrency carry positions, resulting in price movements across both markets. Although this explanation is intuitive, formally testing it falls outside the scope of this study and is left for future research.

5.3.4 Robustness Test

In this section, we conduct a robustness test by including additional risk factors. Table 8 presents the regression results. Panel A shows the results when including the three cryptocurrency factors of Liu, Tsyvinski and Wu (2022) and other factors. We find a persistent, almost decreasing pattern in VIX betas across the carry portfolios. The VIX beta of the high-minus-low portfolio remains

significantly negative, with an estimate of -2.5. The momentum factor is also significant in the high-minus-low returns, although its non-monotonic trend persists across the cross-section portfolios. The explanatory power of this extensive model, as measured by R^2, reaches 7.6%, which is only slightly higher than the 5.6% from the parsimonious model that includes only VIX and the three cryptocurrency factors. Panel C shows the results when including the five cryptocurrency factors of Cong, Karolyi, Tang and Zhao (2022) and other factors.[19] We find that the value and network factors emerge as significant in explaining cryptocurrency carry returns. Notably, the network factor's explanatory power is concentrated among high-interest-rate cryptocurrencies, potentially classifying them as "platform tokens" (Cong et al., 2022). However, even after controlling for these factors, the downward pattern of VIX betas across the Low, Mid, and High portfolios persists and the coefficient of VIX for the HML portfolio remains negative, albeit not statistically significant.

Panels B and D show the result when we decompose the VIX into realized volatility and the variance risk premium. Whether we include the three cryptocurrency factors of Liu, Tsyvinski and Wu (2022) (Panel B) or the five cryptocurrency factors of Cong, Karolyi, Tang and Zhao (2022) (Panel D), the coefficient of realized volatility remains significantly negative. This finding leads us to conclude that the crypto-carry return is substantially influenced by equity volatility risk, even though the C-5 factors provide partial explanations.

[Insert Table 8 Here]

6 Carry Trade between Stablecoins and the U.S. Dollar

This section aims to investigate the carry trade returns using stablecoins, which are excluded in the previous sections. To take advantage of the interest rate differentials, investors could implement a carry trade strategy of borrowing the U.S. dollar, exchanging for stablecoins pegged to the U.S. dollar, and depositing the stablecoins on online platforms to earn a profit, as the interest rates of stablecoins are usually higher than those of the U.S. dollar. This constitutes a carry trade between stablecoins and the U.S. dollar. Notably, this strategy has a unique feature because it focuses solely on profiting from the interest rate spread, without betting on the changes in cryptocurrency values. In this section, we investigate the performance of this strategy.

[19] We thank Weiyi Zhao for kindly sharing the data with us.

The carry trade strategy as described focuses on two stablecoins, USDT (Tether) and DAI, both of which are pegged to the U.S. dollar. The direction of the trade depends on whether the interest rate of the U.S. dollar is higher or lower than the lending and deposit rate of the stablecoins. When the interest rate of borrowing the U.S. dollar is lower than the interest rate of depositing a stablecoin, the strategy involves borrowing the U.S. dollar, exchanging it for a stablecoin, and depositing it on the Bitfinex platform for a specific period of time, such as one month or one week. If the interest rate of depositing the U.S. dollar is higher than the interest rate of borrowing the stablecoin, we execute the trade in reverse by borrowing the stablecoin, exchanging it for U.S. dollars, and depositing the U.S. dollars.[20] If the interest rate of the U.S. dollar falls between the two, no trade is conducted. We form the stablecoin carry portfolio for each day and rebalance the portfolio on a monthly basis. The resulting carry returns are therefore monthly returns at daily frequency.

The return of the strategy for each situation is calculated as follows:

$$r_{t+1} = \begin{cases} i_t^{c-dep} - i_t^{USD} + \Delta s_{t+1} & i_t^{c-dep} > i_t^{USD} \\ i_t^{USD} - i_t^{c-borw} - \Delta s_{t+1} & i_t^{USD} > i_t^{c-borw} \\ 0 & i_t^{c-dep} \leq i_t^{USD} \leq i_t^{c-borw} \end{cases}$$

where i_t^{c-dep} and i_t^{c-borw} are the deposit rate and borrowing rate of a stablecoin at time t, respectively. Δs_{t+1} is the change in log price of a stablecoin from time t to time $t+1$.

Table 9 shows the results.[21] The monthly carry return for both USDT and DAI are significantly positive at the 1% level. As expected, almost all carry returns come from the interest rate differential, since the peg between the U.S. dollar and stablecoins is largely maintained during the sample period. This strategy offers a very high Sharpe ratio. For example, the annualized carry return of USDT calculated from the monthly return is 9.28%, and the Sharpe ratio is as high as 3.54. The result confirms the profitability of the carry strategy between stablecoins and the U.S. dollar.

It is worth noting that the stability of the peg is crucial for the superior excess returns that

[20] We assume that the interest rates of depositing and borrowing the U.S. dollar are the same.

[21] The means, standard deviations, and Newey-West standard errors (Newey and West, 1987) are annualized and in percentage points. The Sharpe ratios are annualized and in real numbers.

stablecoins offer. This means that exploiting the interest rate spread between stablecoins and the U.S. dollar entails some risk, as the strategy depends on the likelihood of the peg being sustained. Gorton, Klee, Ross, Ross and Vardoulakis (2024) show that stablecoins can maintain the peg because they facilitate cryptocurrency speculation. But they could break down if there is a huge shock in the speculation demand. d'Avernas, Maurin and Vandeweyer (2023) analyze the vulnerability of different pegging mechanisms for stablecoins. The risk-return trade-off of the stablecoin carry trade is appealing within the sample period, but it may not be representative of the true depegging risk, as no depeg events occurred during that time. A similar case is the fixed exchange rate regime of El Salvador in the 1990s. During the peg, the colón-dollar interest rate spread averaged over five percentage points, with substantial variation over time. Several factors that typically affect depegging risk, such as net international reserves and fiscal sustainability, are empirically related to the variation in the colón-dollar interest rate spread (e.g., Swiston, 2011). Although we cannot formally test whether the carry trade return compensates for the depegging risk of stablecoins, there is anecdotal evidence that supports this hypothesis.

[Insert Table 9 Here]

The peg of USDT is backed by the physical U.S. dollar reserves. The strength of the peg relies on the adequacy of the reserves. On the other hand, DAI is collateralized by other cryptocurrencies, and the peg is regulated by an algorithmic Ethereum-based protocol. Historically, the value of USDT has deviated less from the parity than DAI (an average tracking error of 0.1% for USDT versus 0.8% for DAI in the post-July 2020 sample). This is in line with the higher carry trade returns in DAI. In other words, the higher carry trade returns of DAI come with higher deviations from parity.

7 Conclusion

Cryptocurrency investment is highly speculative, characterized by high returns and volatility. In this paper, we study the cryptocurrency carry trade—similar to the conventional fiat currency trade strategy of buying high-interest-rate currencies and shorting low-interest-rate currencies, but exclusively within the cryptocurrency universe. We find that the cross-sectional carry trade of cryptocurrencies generates significant positive returns. Compared to the equal-weighted passive

buy-and-hold strategy of all cryptocurrencies, the cryptocurrency carry trade has a higher Sharpe ratio despite having lower average returns. Nevertheless, it features more extreme positive and negative returns despite having lower volatility.

We explore various potential explanations for this phenomenon and find that none of the cryptocurrency factors, such as market, size, momentum, volatility risk, downside risk, platform risk, and liquidity risk, can fully account for the cryptocurrency carry trade returns. We also find that the fiat currency carry factor cannot explain the cryptocurrency carry trade returns. On the other hand, the equity volatility risk factor, as proposed by Lustig, Roussanov and Verdelhan (2011), seems to play a role in generating the positive carry trade returns. Our results highlight the linkages between equity market risk factors and cryptocurrency returns, and suggest that further research is needed to fully understand the determinants of cryptocurrency risk premium.

It is essential to emphasize that while our examination of the cryptocurrency carry trade is approached from the perspective of asset pricing, other channels, especially those related to the decentralized blockchain trading rule, may also play an important role in the cryptocurrency carry trade. These factors include the blockchain protocols (Cong, He and Li, 2021) and wash trading (Cong, Li, Tang and Yang, 2023), among others. Due to constraints in the scope of this paper, we refrain from exploring these additional channels, deferring them to future research.

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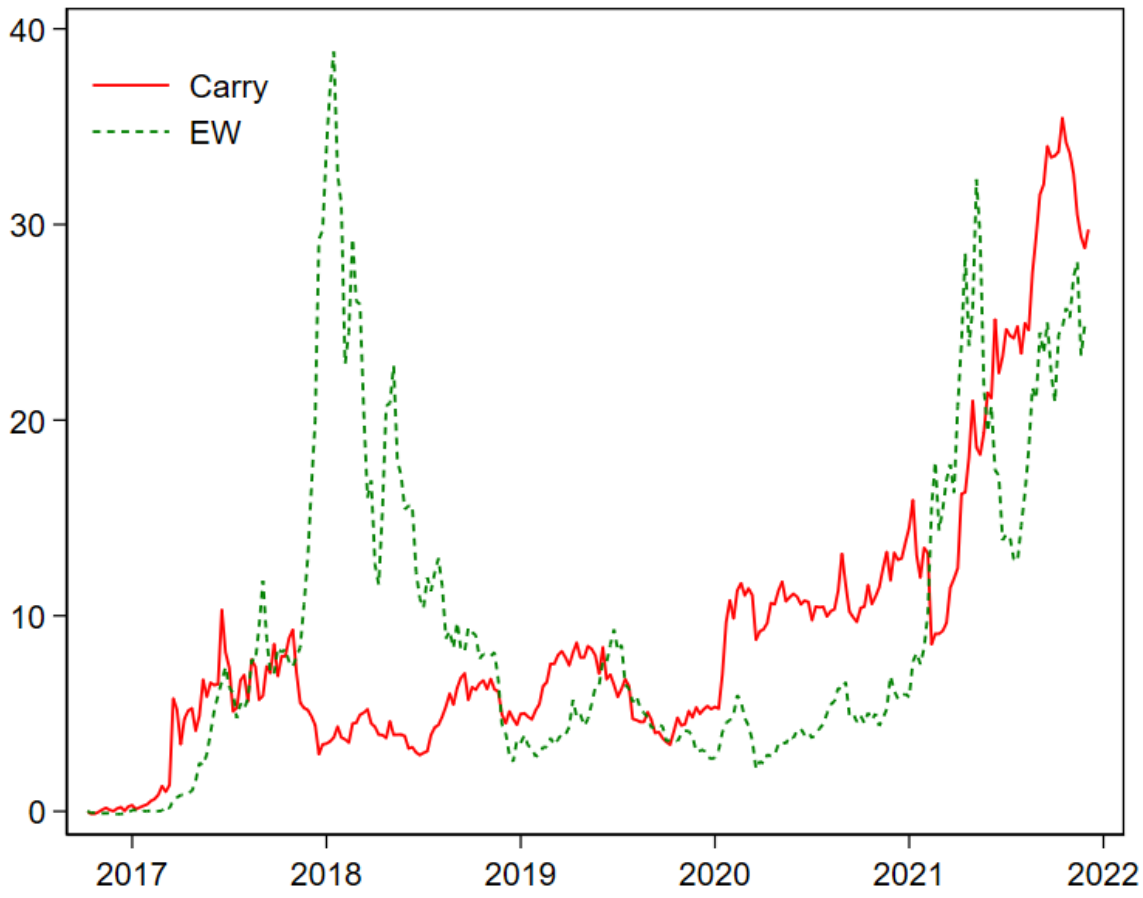
Figure 1: Cumulative Returns of Carry Portfolio and Passive Equal-weighted Portfolio



This figure shows the comparison between the carry portfolio and the passive equal-weighted (EW) portfolio in terms of their cumulative returns. The carry portfolio is formed as in Section 4.1, and is adjusted to match the volatility level of the EW portfolio. The cumulative returns (y-axis) are expressed as real numbers, not percentages. The cumulative return of the carry portfolio up to week T is calculated as: $cmr_{carry,T} = \exp[\Sigma_{t=0}^{T}(r_{High,t} - r_{Low,t})] - 1$, where $r_{High,t}$ and $r_{Low,t}$ denote the log excess returns of High portfolio and Low portfolio at week t, respectively. The figure covers the sample period from October 2016 to November 2021.

Table 1: Summary Statistics of Cryptocurrency Interest Rates and Price Changes

This table provides summary statistics of all cryptocurrencies in our sample. Panel A shows the summary statistics of annualized interest rates. The data are from the Bitfinex platform. Panel B reports summary statistics of daily changes in log prices ($logP_2 - logP_1$) of the cryptocurrencies. We use daily close prices. The data is from the website coinmarketcap.com. The sample period is October 2016 to November 2021. Mean and standard deviations are annualized.

Panel A Summary Statistics of Interest Rates

Currency	0x	Algorand	Avalanche	Axie Infinity	Bitcoin	Bitcoin Cash	Bitcoin Gold	Bitcoin SV	Cardano	Chainlink	Compound	Cosmos
Start	17-Oct-18	3-Mar-20	5-Aug-21	23-Oct-21	1-Oct-16	17-Feb-21	27-Dec-17	21-Dec-18	3-Sep-20	25-Aug-20	10-Nov-21	26-Feb-20
Mean	6.06%	7.56%	22.67%	178.70%	8.26%	0.85%	17.48%	7.80%	17.41%	4.76%	124.39%	6.83%
Std	0.45%	0.60%	1.30%	2.56%	0.50%	0.28%	1.15%	0.41%	2.03%	0.59%	4.01%	0.68%
Min	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	36.50%	0.00%
P25	1.73%	0.63%	4.36%	146.77%	2.05%	0.00%	3.32%	2.35%	0.47%	0.06%	47.42%	0.13%
Median	3.36%	3.86%	11.72%	193.00%	4.95%	0.00%	9.03%	5.78%	1.61%	1.15%	129.22%	0.93%
P75	6.79%	9.44%	36.50%	210.97%	9.47%	0.00%	23.25%	9.55%	7.07%	3.35%	198.42%	10.53%
Max	55.45%	95.07%	113.34%	248.20%	45.83%	46.40%	126.43%	40.23%	158.80%	77.27%	211.09%	70.72%
Skewness	321.19%	321.97%	170.38%	-137.53%	201.68%	681.74%	228.13%	182.24%	254.88%	422.74%	-5.41%	291.50%
Kurtosis	1508.45%	1697.08%	589.60%	595.20%	673.83%	4948.67%	915.22%	655.83%	815.34%	2352.39%	113.27%	1199.39%

Currency	Dai	Dash	Dogecoin	EOS	Elrond	Ethereum	Ethereum Classic	FTX Token	Fantom	Filecoin	IOTA	LUNA
Start	23-Jul-20	16-Mar-17	4-May-21	8-Jul-17	28-Oct-21	1-Oct-16	1-Oct-16	6-Mar-20	22-Oct-21	23-Nov-20	3-Jul-17	5-Aug-21
Mean	13.89%	8.12%	10.46%	6.73%	160.09%	6.04%	9.70%	26.98%	82.00%	50.72%	1.12%	58.90%
Std	1.67%	0.59%	1.07%	0.80%	3.12%	0.29%	0.83%	2.11%	3.48%	3.11%	0.14%	3.22%
Min	0.00%	0.00%	0.31%	0.00%	0.00%	0.14%	0.00%	0.00%	36.50%	0.00%	0.00%	3.56%
P25	0.56%	1.47%	0.66%	0.13%	155.30%	2.01%	0.67%	0.48%	44.73%	7.22%	0.00%	6.93%
Median	5.33%	4.06%	1.87%	0.75%	178.79%	4.21%	2.90%	6.11%	53.31%	26.07%	0.26%	35.80%
P75	7.97%	10.64%	8.03%	5.43%	191.57%	8.05%	11.79%	37.78%	78.25%	73.07%	0.80%	96.58%
Max	157.92%	87.57%	108.34%	93.45%	238.03%	28.11%	96.19%	175.32%	252.96%	227.42%	19.18%	215.53%
Skewness	321.79%	306.28%	278.69%	382.00%	-146.66%	148.72%	283.34%	170.88%	187.14%	145.56%	424.67%	101.69%
Kurtosis	1244.49%	1488.32%	1064.73%	1880.15%	449.92%	487.35%	1248.15%	503.96%	497.23%	421.96%	2341.44%	280.55%

Currency	Litecoin	Maker	Metaverse ETP	Monero	Neo	OMG Network	Polkadot	Polygon	SHIBA INU	Santiment Network	Solana	Stellar
Start	1-Oct-16	9-Nov-21	20-Oct-17	14-Mar-17	18-Sep-17	4-Aug-17	2-Sep-20	23-Oct-21	11-Nov-21	18-Sep-17	31-May-21	18-Aug-19
Mean	6.22%	30.47%	11.84%	5.23%	13.95%	6.05%	6.15%	166.99%	25.52%	9.68%	19.61%	5.41%
Std	0.50%	0.82%	1.53%	0.38%	0.98%	0.57%	1.29%	3.39%	1.68%	1.44%	1.66%	0.70%
Min	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	36.39%	2.95%	0.00%	0.05%	0.00%
P25	1.17%	36.50%	0.00%	1.20%	2.23%	0.27%	0.00%	109.51%	7.21%	0.00%	1.80%	0.13%
Median	3.20%	36.50%	0.95%	2.79%	6.48%	1.66%	0.07%	206.66%	8.84%	0.34%	8.46%	0.68%
P75	7.12%	36.50%	7.58%	6.55%	17.36%	5.78%	0.33%	215.10%	28.78%	5.69%	25.39%	3.08%
Max	65.41%	54.09%	188.38%	47.83%	103.87%	59.29%	158.14%	236.24%	114.88%	222.49%	197.73%	90.55%
Skewness	359.18%	-129.44%	399.13%	333.76%	228.39%	276.20%	495.73%	-71.08%	189.80%	524.65%	342.56%	390.07%
Kurtosis	1848.23%	333.83%	2043.32%	1694.55%	855.04%	1078.27%	2733.30%	206.35%	576.14%	3477.87%	1659.02%	1959.94%

Currency	SushiSwap	TRON	Tether	Tezos	Uniswap	Unus Sed LEO	XRP	Zcash	pNetwork	yearn.finance
Start	20-Jan-21	2-Sep-20	21-Dec-18	15-Aug-19	30-Sep-20	19-Jun-19	4-Jul-17	20-Dec-16	8-Jul-20	11-Nov-20
Mean	24.04%	11.04%	12.03%	10.79%	13.00%	0.01%	3.06%	4.80%	8.10%	23.72%
Std	1.30%	0.93%	0.47%	0.81%	0.92%	0.00%	0.31%	0.48%	0.75%	1.45%
Min	1.38%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	2.08%
P25	8.41%	0.00%	6.28%	0.40%	0.62%	0.00%	0.13%	0.26%	0.00%	3.06%
Median	15.24%	2.00%	10.93%	5.33%	3.73%	0.00%	0.67%	1.33%	0.62%	14.11%
P75	28.86%	15.11%	14.59%	12.15%	25.55%	0.00%	3.07%	4.38%	11.59%	37.70%
Max	120.94%	85.22%	51.56%	76.83%	101.38%	1.07%	45.95%	56.80%	64.77%	167.18%
Skewness	211.86%	195.52%	188.88%	221.29%	195.41%	1355.76%	380.53%	311.61%	227.03%	218.43%
Kurtosis	736.03%	624.70%	784.74%	761.25%	779.57%	19069.91%	2052.64%	1331.28%	781.65%	952.42%

Panel B Summary Statistics of Daily Price Changes

Currency	0x	Algorand	Avalanche	Axie Infinity	Bitcoin	Bitcoin Cash	Bitcoin Gold	Bitcoin SV	Cardano	Chainlink	Compound	Cosmos
Mean	0.02%	0.25%	1.81%	0.23%	0.24%	-0.08%	-0.11%	0.03%	0.61%	0.13%	-0.81%	0.31%
Std	6.17%	7.15%	8.41%	4.56%	4.15%	6.66%	6.47%	6.67%	6.32%	6.99%	4.76%	7.19%
Min	-46.01%	-57.99%	-22.40%	-10.56%	-46.47%	-43.46%	-37.15%	-56.03%	-30.11%	-46.60%	-9.08%	-49.29%
P25	-3.22%	-3.35%	-2.40%	-2.20%	-1.45%	-3.01%	-2.76%	-2.26%	-2.53%	-3.73%	-4.02%	-3.05%
Median	0.00%	0.00%	1.49%	0.57%	0.23%	0.48%	0.00%	0.00%	0.13%	0.32%	-1.65%	0.08%
P75	3.15%	3.74%	5.83%	3.13%	2.07%	2.90%	2.43%	2.13%	3.12%	4.08%	2.74%	3.68%
Max	44.04%	41.33%	24.64%	12.07%	22.51%	42.08%	71.66%	58.45%	27.94%	27.56%	8.29%	28.00%
Skewness	9.94%	-64.64%	14.61%	-9.62%	-77.45%	10.23%	140.55%	83.80%	36.92%	-65.62%	14.64%	-54.02%
Kurtosis	1050.43%	1303.56%	411.10%	334.92%	1425.82%	1577.27%	2133.56%	2471.21%	591.93%	776.56%	212.96%	981.27%

Currency	Dai	Dash	Dogecoin	EOS	Elrond	Ethereum	Ethereum Classic	FTX Token	Fantom	Filecoin	IOTA	LUNA
Mean	-0.01%	0.04%	-0.44%	0.03%	1.16%	0.31%	0.20%	0.46%	-0.51%	0.16%	0.08%	1.08%
Std	0.38%	6.19%	7.72%	6.97%	5.81%	5.55%	6.42%	5.17%	7.88%	7.35%	6.90%	8.08%
Min	-1.96%	-46.55%	-45.23%	-50.32%	-12.66%	-55.07%	-50.78%	-32.18%	-18.94%	-42.62%	-50.56%	-19.13%
P25	0.00%	-2.58%	-3.33%	-2.58%	-2.01%	-2.03%	-2.30%	-2.06%	-5.15%	-2.58%	-3.04%	-4.17%
Median	0.00%	0.00%	-0.12%	0.00%	-0.50%	0.08%	0.00%	0.33%	-1.43%	-0.29%	0.00%	0.00%
P75	0.00%	2.62%	2.17%	2.63%	3.32%	2.76%	2.52%	3.01%	3.32%	2.60%	3.36%	5.73%
Max	1.96%	45.13%	24.09%	43.94%	13.81%	29.01%	45.77%	29.56%	20.02%	45.87%	38.40%	35.56%
Skewness	-1.55%	28.12%	-98.47%	-7.79%	48.49%	-43.28%	10.03%	6.26%	41.26%	70.84%	5.88%	100.90%
Kurtosis	1340.32%	1262.84%	1114.33%	1081.29%	327.66%	1199.87%	1163.99%	910.45%	376.43%	1299.29%	847.57%	577.23%

Currency	Litecoin	Maker	Metaverse ETP	Monero	Neo	OMG Network	Polkadot	Polygon	SHIBA INU	Santiment Network	Solana	Stellar
Mean	0.21%	-0.12%	-0.19%	0.14%	0.04%	0.07%	0.40%	0.23%	-0.93%	0.02%	1.01%	0.19%
Std	5.92%	4.50%	8.34%	5.88%	6.35%	7.31%	7.41%	5.92%	6.84%	7.51%	6.67%	5.97%
Min	-44.90%	-7.26%	-101.16%	-53.42%	-45.39%	-45.38%	-47.70%	-9.36%	-12.59%	-44.33%	-28.13%	-36.00%
P25	-2.40%	-3.56%	-3.39%	-2.41%	-2.85%	-3.39%	-3.36%	-4.01%	-3.75%	-3.46%	-3.16%	-2.52%
Median	0.00%	-0.31%	-0.25%	0.06%	0.00%	0.00%	0.00%	-1.05%	-1.37%	0.00%	0.65%	0.02%
P75	2.56%	1.24%	3.03%	2.93%	3.14%	3.36%	3.69%	3.68%	2.18%	3.32%	4.58%	2.43%
Max	51.04%	8.36%	105.99%	43.03%	34.44%	54.17%	31.94%	19.08%	15.18%	39.05%	19.77%	55.93%
Skewness	32.25%	43.58%	15.15%	-44.13%	-25.17%	44.03%	-12.01%	86.96%	80.85%	-5.55%	-5.67%	120.71%
Kurtosis	1373.30%	225.32%	3882.12%	1420.43%	876.05%	981.58%	871.95%	417.95%	374.79%	844.36%	458.37%	1797.57%

Currency	SushiSwap	TRON	Tether	Tezos	Uniswap	Unus Sed LEO	XRP	Zcash	pNetwork	yearn.finance
Mean	0.03%	0.23%	0.00%	0.18%	0.39%	0.07%	0.09%	0.10%	0.10%	0.13%
Std	7.97%	6.26%	0.41%	6.57%	7.60%	2.90%	6.48%	6.27%	9.03%	6.64%
Min	-23.84%	-38.34%	-2.03%	-60.55%	-40.33%	-19.96%	-55.04%	-53.94%	-41.43%	-33.51%
P25	-4.76%	-2.62%	0.00%	-2.82%	-3.91%	-0.84%	-2.33%	-2.94%	-4.62%	-2.78%
Median	-0.36%	0.34%	0.00%	0.00%	0.26%	0.00%	-0.01%	0.00%	0.00%	0.42%
P75	4.07%	2.83%	0.00%	3.06%	4.39%	0.95%	2.05%	3.14%	3.70%	3.24%
Max	28.21%	33.41%	1.98%	25.05%	38.02%	26.08%	60.69%	52.82%	68.59%	38.67%
Skewness	31.34%	-14.65%	-10.35%	-124.41%	39.34%	56.14%	93.43%	2.68%	130.91%	25.94%
Kurtosis	387.49%	911.07%	861.89%	1539.64%	786.73%	1885.50%	1846.28%	1091.64%	1280.38%	875.43%

Table 2: Predicting cryptocurrency excess returns using carry

This table reports the results of panel regression to test the predictive power of carry with all 44 cryptocurrencies in our sample (stablecoins are not included). The regression equation is $r_{j,t+1} = \alpha_j + \beta(i^c_{j,t} - i^{USD}_t) + \gamma_t + \epsilon_{j,t+1}$, where α_j denotes individual fixed effects and γ_t represents time fixed effects. Newey-West standard errors with four lags are reported in parentheses. ***, **, and * indicate $p < 0.01$, $p < 0.05$, and $p < 0.1$, respectively. The sample period is October 2016 to November 2021.

	Individual fixed effects	Time fixed effects	β	N
Weekly return	X	X	1.66**	24,111
			(0.72)	
	X		2.11**	24,111
			(0.96)	
		X	1.68***	24,111
			(0.58)	
Monthly return	X	X	1.69***	14,379
			(0.50)	
	X		1.58***	14,379
			(0.68)	
		X	1.69***	14,379
			(0.46)	

Table 3: Top Five Most Frequently Occurring Cryptocurrencies in the High and Low Portfolios

This table shows the top five most frequently occurring cryptocurrencies in the High and Low portfolios. It summaries the start and end of the sample period, total number of weeks in the sample, frequencies appearing in the Low, Mid, and High portfolios, and market capitalization for each cryptocurrency. The market capitalization is in billion U.S. dollars.

	Start	End	Weeks	Low	Mid	High	Market Cap.
Panel A: Top Five Most Frequently Occurring Cryptocurrencies in the High Portfolio							
Bitcoin Gold	12-Jan-18	30-Nov-21	203	16	53	134	0.56
Bitcoin	20-Oct-16	24-Nov-21	266	42	100	124	261.45
Neo	29-Sep-17	30-Nov-21	218	28	67	123	2.05
Ethereum Classic	13-Oct-16	30-Nov-21	268	105	71	92	1.78
Ethereum	13-Oct-16	30-Nov-21	268	39	144	85	80.61
Panel B: Top Five Most Frequently Occurring Cryptocurrencies in the Low Portfolio							
IOTA	14-Jul-17	30-Nov-21	229	175	47	7	2.16
Santiment Network Token	29-Sep-17	30-Nov-21	218	133	40	45	0.04
Unus Sed LEO	1-Jul-19	30-Nov-21	126	126	0	0	1.62
Zcash	30-Dec-16	30-Nov-21	257	118	99	40	0.79
EOS	21-Jul-17	30-Nov-21	228	117	59	52	4.02

Table 4: Summary Statistics of Carry Trade Portfolios

This table presents summary statistics of carry portfolios. Low (Mid, High) represents the portfolio containing tokens with the lowest (middle, highest) interest rates. HmL represents the long-short portfolio that buys tokens in the High portfolio and shorts those in the Low portfolio. EW represents a portfolio constructed as the equal-weighted average of all cryptocurrencies in our sample. Portfolios are rebalanced weekly. Average return, return on Δs, and return on interest differential and corresponding standard errors are annualized and in percentage points. We multiply the weekly mean by 52 and weekly standard deviation (volatility) by $\sqrt{52}$ to calculate the annualized mean and standard deviation. This table also summarizes characteristics of carry trade portfolios, including age, market capitalization, and trading volume. For cryptocurrencies launched at or earlier than start of the sample (October 2016), we define their age as one week at that time. To calculate the average market capitalization of each group, we firstly calculate the average market capitalization of all cryptocurrencies on rebalance days in each group, and then calculate the time-series average for each group. Similarly, to calculate the average trading volume of each group, we first calculate the average trading volume of all cryptocurrencies on re-balance days in each group, and then calculate the time-series average for each group. Newey-West standard errors with four lags are reported in parentheses. *, **, and *** represent statistical significance at the 10%, 5%, and 1% levels, respectively. The sample period is October 2016 to November 2021.

	Low	Mid	High	HmL	EW
Average return	50.4	56.1	93.9*	43.4*	63.1
	(48.9)	(48.2)	(54.4)	(24.0)	(48.5)
Return on interest rate differential	-0.3	2.8***	19.1***	19.4***	6.7***
	(0.2)	(0.5)	(1.3)	(1.3)	(0.6)
Return on change in prices	50.7	53.3	74.7	24.0	56.4
	(48.9)	(48.1)	(54.3)	(24.1)	(48.4)
Vol	93.8	92.6	102.7	58.9	90.1
Skewness	0.15	0.06	0.63	1.95	-0.09
Kurtosis	4.3	3.4	8.3	21.1	3.7
Sharpe ratio	0.54	0.61	0.91	0.74	0.70
# obs.	268	268	268	268	268
Age (weeks)	77.5	88.1	74.9		80.2
Market cap (billion U.S. dollars)	10.5	21.1	19.1		16.6
Trading volume (billion U.S. dollars)	0.96	2.28	2.03		1.74

Table 5: Carry Trade Returns and Risk Factors

This table reports the regression results of cryptocurrency carry returns on different risk factors. Low (Mid, High) represents the portfolio containing tokens with the lowest (middle, highest) interest rates. HmL represents the long-short portfolio that buys tokens in the High portfolio and shorts those in the Low portfolio. Panels A-D and F report the regression results on cryptocurrency three factors, five factors, market volatility, downside risk, and illiquidity risk, respectively. Panel E reports the cumulative abnormal returns of carry portfolios during events related to cryptocurrency platform collapses. Panel G reports the regression results on fiat-currency carry factor. Robust standard errors are reported in parentheses in Panel E, and Newey-West standard errors with four lags are reported for all other panels. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively. The data are weekly and cover the period from October 2016 to November 2021.

Panel A: Regression on Cryptocurrency Three Factors

	Low	Mid	High	HmL
Constant	-0.012***	-0.010**	-0.003	0.009**
	(0.004)	(0.004)	(0.006)	(0.004)
Cmkt	0.831***	0.865***	0.800***	-0.031
	(0.058)	(0.044)	(0.071)	(0.056)
Csize	0.062	0.039	0.019	-0.043
	(0.041)	(0.038)	(0.042)	(0.026)
Cmom	-0.128	-0.158*	0.032	0.160*
	(0.096)	(0.091)	(0.149)	(0.091)
R^2	0.597	0.655	0.454	0.045

Panel B: Regression on Cryptocurrency Five Factors

	Low	Mid	High	HmL
Constant	-0.005	-0.002	-0.002	0.003
	(0.006)	(0.006)	(0.007)	(0.004)
Mkt	0.701***	0.658***	0.712***	0.011
	(0.067)	(0.070)	(0.070)	(0.045)
Size	-0.008	-0.017	-0.006	0.002
	(0.014)	(0.016)	(0.017)	(0.011)
Momentum	-0.014	-0.026	0.109	0.123
	(0.090)	(0.076)	(0.127)	(0.087)
Value	-0.049	-0.035	0.046	0.096***
	(0.063)	(0.057)	(0.062)	(0.032)
Network	-0.035	0.006	0.133*	0.168***
	(0.077)	(0.064)	(0.079)	(0.050)
R^2	0.391	0.357	0.383	0.107

Panel C: Regression on Cryptocurrency Volatility

	Low		Mid		High		HmL	
Constant	0.010	-0.013***	0.011	-0.010**	0.018*	-0.004	0.009*	0.010**
	(0.009)	(0.004)	(0.009)	(0.004)	(0.010)	(0.006)	(0.005)	(0.004)
$\Delta\sigma^{\text{crypto}}$	-0.616	0.683	-0.960*	0.317	-0.332	0.845	0.284	0.162
	(0.634)	(0.463)	(0.547)	(0.302)	(0.778)	(0.644)	(0.416)	(0.422)
Cmkt		0.849***		0.873***		0.821***		-0.027
		(0.060)		(0.044)		(0.067)		(0.055)
Csize		0.075		0.043		0.033		-0.042
		(0.046)		(0.039)		(0.049)		(0.029)
Cmom		-0.137		-0.148		0.037		0.174*
		(0.096)		(0.092)		(0.147)		(0.090)
R^2	0.007	0.606	0.017	0.658	0.002	0.465	0.004	0.052

Panel D: Regression on Cryptocurrency Downside Risk								
	Low		Mid		High		HmL	
Constant	0.056***	0.004	0.056***	-0.005	0.065***	0.021**	0.009*	0.017**
	(0.008)	(0.006)	(0.008)	(0.006)	(0.008)	(0.009)	(0.005)	(0.008)
Cds	-1.417***	-0.382***	-1.361***	-0.119	-1.428***	-0.559***	-0.011	-0.177
	(0.094)	(0.130)	(0.108)	(0.097)	(0.114)	(0.153)	(0.078)	(0.142)
Cmkt		0.677***		0.817***		0.574***		-0.102
		(0.098)		(0.058)		(0.090)		(0.091)
Csize		0.066*		0.040		0.026		-0.041
		(0.038)		(0.037)		(0.041)		(0.027)
Cmom		-0.110		-0.153*		0.057		0.168*
		(0.091)		(0.092)		(0.145)		(0.093)
R^2	0.445	0.610	0.420	0.657	0.377	0.477	0.000	0.052

Panel E: Platform Risk (Event Study)				
	Low	Mid	High	HmL
CAR(-1,+1)	-0.069	-0.107**	-0.073	-0.004
	(0.046)	(0.036)	(0.066)	(0.037)
CAR(-2,+2)	-0.013	-0.067	0.044	0.057
	(0.053)	(0.049)	(0.061)	(0.041)
CAR(-4,+4)	-0.047	-0.043	-0.044	0.003
	(0.071)	(0.077)	(0.177)	(0.159)

Panel F: Regression on Cryptocurrency Illiquidity Risk								
	Low		Mid		High		HmL	
Constant	0.011	-0.011**	0.010	-0.011**	0.019*	-0.002	0.008	0.009*
	(0.010)	(0.005)	(0.010)	(0.005)	(0.011)	(0.007)	(0.005)	(0.005)
Illiquidity	-4.588	-3.979	3.474	3.584	-4.481	-4.217	0.108	-0.238
	(9.089)	(5.382)	(9.269)	(4.097)	(10.486)	(5.991)	(3.240)	(3.131)
Cmkt		0.831***		0.865***		0.800***		-0.031
		(0.058)		(0.044)		(0.072)		(0.056)
Csize		0.060		0.040		0.017		-0.043
		(0.041)		(0.038)		(0.042)		(0.026)
Cmom		-0.128		-0.158*		0.032		0.160*
		(0.096)		(0.092)		(0.149)		(0.091)
R^2	0.001	0.598	0.000	0.656	0.000	0.455	0.000	0.045

Panel G: Regression on Fiat Currency Carry Factor								
	Low		Mid		High		HmL	
Constant	0.010	-0.014***	0.013	-0.010**	0.017*	-0.005	0.008*	0.009**
	(0.010)	(0.004)	(0.010)	(0.004)	(0.010)	(0.006)	(0.004)	(0.004)
Fiat Carry	-0.099	0.894*	-1.098	-0.091	0.329	1.075*	0.428	0.181
	(0.784)	(0.506)	(0.820)	(0.417)	(0.867)	(0.606)	(0.451)	(0.436)
Cmkt		0.836***		0.864***		0.806***		-0.030
		(0.057)		(0.044)		(0.071)		(0.056)
Csize		0.065*		0.038		0.023		-0.042
		(0.038)		(0.038)		(0.041)		(0.027)
Cmom		-0.137		-0.157*		0.021		0.158*
		(0.095)		(0.091)		(0.148)		(0.091)
R^2	0.000	0.602	0.007	0.655	0.000	0.459	0.003	0.045

Table 6: Carry Trade Returns and VIX

This table reports the regression results of carry portfolio returns on change in VIX. Panel A controls for cryptocurrency three factors from Liu et al. (2022) and Panel B controls for cryptocurrency five factors from Cong et al. (2022). The daily data of VIX are obtained from the CBOE, and we calculate the weekly change in VIX. Change in VIX is scaled by a factor of 0.001. Low (Mid, High) represents the portfolio containing tokens with the lowest (middle, highest) interest rates. HmL represents the long-short portfolio that buys tokens in the High portfolio and shorts those in the Low portfolio. Portfolios are rebalanced weekly. The sample period spans from October 2016 to November 2021. Newey-West standard errors with four lags are reported in parentheses. *, **, and *** represent statistical significance at the 10%, 5%, and 1% levels, respectively.

Panel A: Regression on Change in VIX Controlling for C3 Factors								
	Low		Mid		High		HmL	
Constant	0.010	-0.012***	0.011	-0.010**	0.018*	-0.002	0.008*	0.010**
	(0.009)	(0.004)	(0.009)	(0.004)	(0.010)	(0.006)	(0.005)	(0.004)
ΔVIX	-6.207***	-0.317	-6.233***	-0.040	-8.068***	-2.623	-1.861*	-2.306*
	(2.333)	(1.287)	(1.820)	(1.095)	(2.858)	(1.661)	(0.959)	(1.201)
Cmkt		0.828***		0.865***		0.779***		-0.049
		(0.061)		(0.044)		(0.072)		(0.056)
Csize		0.062		0.039		0.022		-0.040
		(0.040)		(0.038)		(0.040)		(0.029)
Cmom		-0.127		-0.158*		0.037		0.164*
		(0.096)		(0.092)		(0.149)		(0.092)
R^2	0.034	0.598	0.035	0.655	0.048	0.459	0.008	0.056

Panel B: Regression on Change in VIX Controlling for C5 Factors								
	Low		Mid		High		HmL	
Constant	0.010	-0.004	0.011	-0.002	0.018*	-0.000	0.008*	0.004
	(0.009)	(0.006)	(0.009)	(0.006)	(0.010)	(0.006)	(0.005)	(0.004)
ΔVIX	-6.207***	-2.295	-6.233***	-2.511*	-8.068***	-3.969**	-1.861*	-1.674*
	(2.333)	(1.558)	(1.820)	(1.302)	(2.858)	(1.783)	(0.959)	(0.956)
Mkt		0.685***		0.640***		0.683***		-0.001
		(0.066)		(0.071)		(0.064)		(0.043)
Size		-0.009		-0.018		-0.008		0.001
		(0.014)		(0.016)		(0.017)		(0.011)
Momentum		-0.009		-0.020		0.118		0.127
		(0.088)		(0.075)		(0.125)		(0.088)
Value		-0.055		-0.042		0.036		0.091***
		(0.064)		(0.057)		(0.063)		(0.032)
Network		-0.038		0.003		0.127		0.165***
		(0.077)		(0.064)		(0.079)		(0.050)
R^2	0.034	0.395	0.035	0.363	0.048	0.394	0.008	0.113

Table 7: Decomposition of VIX: Realized Volatility and Variance Risk Premium

This table reports the regression results of carry portfolio returns on change in realized volatility (ΔRVOL) and variance risk premium (VRP). Panel A and Panel B report the regression results of carry portfolio returns on change in realized volatility, controlling for cryptocurrency three factors from Liu et al. (2022) and cryptocurrency five factors from Cong et al. (2022), respectively. Panel C and Panel D report the regression results of carry portfolio returns on variance risk premium. Panel E and Panel F report the regression results including both realized volatility and variance risk premium. We calculate the variance risk premium as $VRP_t = VIX_t^2 - RVOL_t^2$, where VRP_t and $RVOL_t$ represent the variance risk premium and realized volatility on day t, respectively. The daily data of VIX and realized volatility are obtained from CBOE. Realized volatility is scaled by a factor of 0.001, and variance risk premium is scaled by a factor of 0.01. Low (Mid, High) represents the portfolio containing tokens with the lowest (middle, highest) interest rates. HmL represents the long-short portfolio that buys tokens in the High portfolio and shorts those in the Low portfolio. Portfolios are rebalanced weekly. The sample period spans from October 2016 to November 2021. Newey-West standard errors with four lags are reported in parentheses. *, **, and *** represent statistical significance at the 10%, 5%, and 1% levels, respectively.

Panel A: Regression on ΔRVOL Controlling for C3 Factors

	Low		Mid		High		HML	
Constant	0.010	-0.012***	0.011	-0.010**	0.018*	-0.002	0.008*	0.010**
	(0.009)	(0.004)	(0.009)	(0.004)	(0.010)	(0.006)	(0.005)	(0.004)
ΔRVOL	-4.075	-1.347	-3.406*	-0.559	-6.320**	-3.567**	-2.246***	-2.220**
	(2.682)	(0.883)	(1.930)	(0.989)	(3.151)	(1.494)	(0.736)	(0.913)
cmkt		0.825***		0.863***		0.785***		-0.040
		(0.059)		(0.044)		(0.069)		(0.053)
csize		0.063		0.039		0.022		-0.041
		(0.039)		(0.037)		(0.040)		(0.027)
cmom		-0.129		-0.159*		0.028		0.157*
		(0.095)		(0.092)		(0.149)		(0.092)
R^2	0.016	0.599	0.011	0.656	0.032	0.464	0.012	0.056

Panel B: Regression on ΔRVOL Controlling for C5 Factors

	Low		Mid		High		HML	
Constant	0.010	-0.005	0.011	-0.002	0.018*	-0.001	0.008*	0.003
	(0.009)	(0.005)	(0.009)	(0.006)	(0.010)	(0.006)	(0.005)	(0.004)
ΔRVOL	-4.075	-0.713	-3.406*	-0.249	-6.320**	-3.088	-2.246***	-2.375***
	(2.682)	(1.605)	(1.930)	(1.322)	(3.151)	(1.931)	(0.736)	(0.739)
Mkt		0.697***		0.657***		0.693***		-0.004
		(0.065)		(0.070)		(0.063)		(0.043)
Size		-0.008		-0.017		-0.007		0.002
		(0.014)		(0.016)		(0.017)		(0.011)
Momentum		-0.013		-0.025		0.112		0.126
		(0.090)		(0.076)		(0.126)		(0.087)
Value		-0.049		-0.035		0.046		0.095***
		(0.063)		(0.057)		(0.063)		(0.031)
Network		-0.034		0.006		0.135*		0.169***
		(0.078)		(0.064)		(0.080)		(0.050)
R^2	0.016	0.391	0.011	0.357	0.032	0.390	0.012	0.120

Panel C: Regression on VRP Controlling for C3 Factors								
	Low		Mid		High		HmL	
Constant	0.010	-0.012***	0.011	-0.010**	0.018*	-0.003	0.008*	0.009**
	(0.009)	(0.004)	(0.009)	(0.004)	(0.010)	(0.006)	(0.005)	(0.005)
VRP	0.638	7.112	-7.052*	0.088	-1.247	2.511	-1.885	-4.600*
	(4.450)	(4.498)	(3.861)	(3.139)	(5.608)	(5.919)	(2.177)	(2.538)
Cmkt		0.832***		0.865***		0.800***		-0.032
		(0.058)		(0.044)		(0.072)		(0.056)
Csize		0.061		0.039		0.018		-0.043
		(0.041)		(0.038)		(0.043)		(0.026)
Cmom		-0.133		-0.158*		0.030		0.164*
		(0.096)		(0.093)		(0.151)		(0.092)
R^2	0.000	0.599	0.001	0.655	0.000	0.455	0.000	0.046
Panel D: Regression on VRP Controlling for C5 Factors								
	Low		Mid		High		HmL	
Constant	0.010	-0.005	0.011	-0.002	0.018*	-0.002	0.008*	0.003
	(0.009)	(0.006)	(0.009)	(0.006)	(0.010)	(0.007)	(0.005)	(0.004)
VRP	0.638	1.155	-7.052*	-6.249**	-1.247	0.886	-1.885	-0.269
	(4.450)	(3.626)	(3.861)	(2.852)	(5.608)	(5.013)	(2.177)	(2.578)
Mkt		0.701***		0.658***		0.712***		0.011
		(0.067)		(0.070)		(0.071)		(0.045)
Size		-0.008		-0.016		-0.006		0.002
		(0.014)		(0.016)		(0.017)		(0.011)
Momentum		-0.014		-0.027		0.109		0.123
		(0.090)		(0.077)		(0.127)		(0.087)
Value		-0.049		-0.037		0.047		0.096***
		(0.063)		(0.057)		(0.063)		(0.032)
Network		-0.034		0.005		0.133*		0.168***
		(0.077)		(0.064)		(0.080)		(0.050)
R^2	0.000	0.391	0.001	0.358	0.000	0.383	0.000	0.107
Panel E: Regression on ΔRVOL and VRP Controlling for C3 Factors								
	Low		Mid		High		HmL	
Constant	0.010	-0.012***	0.011	-0.010**	0.018*	-0.002	0.008*	0.010**
	(0.009)	(0.004)	(0.009)	(0.004)	(0.010)	(0.006)	(0.005)	(0.004)
ΔRVOL	-4.087	-1.402*	-3.360*	-0.560	-6.323**	-3.592**	-2.236***	-2.190**
	(2.648)	(0.796)	(2.013)	(0.992)	(3.143)	(1.439)	(0.759)	(0.990)
VRP	1.670	7.463**	-6.203	0.228	0.349	3.411	-1.320	-4.052**
	(4.477)	(3.771)	(4.409)	(2.794)	(4.816)	(4.013)	(1.256)	(2.038)
Cmkt		0.827***		0.863***		0.786***		-0.041
		(0.059)		(0.044)		(0.070)		(0.054)
Csize		0.062		0.039		0.022		-0.041
		(0.040)		(0.037)		(0.040)		(0.027)
Cmom		-0.135		-0.159*		0.025		0.161*
		(0.096)		(0.093)		(0.151)		(0.093)
R^2	0.016	0.601	0.013	0.656	0.032	0.465	0.012	0.058

Panel F: Regression on ΔRVOL and VRP Controlling for C5 Factors								
	Low		Mid		High		HmL	
Constant	0.010	-0.005	0.011	-0.002	0.018*	-0.001	0.008*	0.003
	(0.009)	(0.005)	(0.009)	(0.006)	(0.010)	(0.006)	(0.005)	(0.004)
ΔRVOL	-4.087	-0.723	-3.360*	-0.202	-6.323**	-3.101	-2.236***	-2.378***
	(2.648)	(1.592)	(2.013)	(1.327)	(3.143)	(1.912)	(0.759)	(0.738)
VRP	1.670	1.339	-6.203	-6.198**	0.349	1.674	-1.320	0.335
	(4.477)	(3.441)	(4.409)	(2.805)	(4.816)	(3.589)	(1.256)	(1.571)
Mkt		0.697***		0.657***		0.693***		-0.004
		(0.065)		(0.070)		(0.063)		(0.043)
Size		-0.009		-0.016		-0.007		0.002
		(0.014)		(0.016)		(0.017)		(0.011)
Momentum		-0.013		-0.027		0.113		0.126
		(0.090)		(0.077)		(0.127)		(0.087)
Value		-0.049		-0.037		0.046		0.095***
		(0.063)		(0.057)		(0.063)		(0.031)
Network		-0.034		0.005		0.136*		0.170***
		(0.078)		(0.065)		(0.081)		(0.050)
R^2	0.016	0.391	0.013	0.359	0.032	0.390	0.012	0.120

Table 8: Robustness: Carry Trade Returns and All Factors

This table presents the results of a multivariate regression analysis of carry portfolio returns on all the previously examined factors. In Panel A we include VIX, cryptocurrency three factors from Liu et al. (2022), and other factors. In Panel B we include realized volatility, variance risk premium, cryptocurrency three factors from Liu et al. (2022), and other factors. In Panel C, we include VIX, cryptocurrency five factors from Cong et al. (2022), and other factors. In Panel D we include realized volatility, variance risk premium, cryptocurrency five factors from Cong et al. (2022), and other factors. Low (Mid, High) represents the portfolio containing tokens with the lowest (middle, highest) interest rates. HmL represents the long-short portfolio that buys tokens in the High portfolio and shorts those in the Low portfolio. Portfolios are rebalanced weekly. Newey-West standard errors are reported in parentheses. *, **, and *** represent statistical significance at the 10%, 5%, and 1% levels, respectively. The data are weekly and cover the period from October 2016 to November 2021. The change in VIX, change in realized volatility, variance risk premium, and change in geopolitical risk are scaled by factors of 0.001, 0.001, 0.01, and 0.0001, respectively, in the regression. This ensures that the coefficients have reasonable magnitudes that are easy to present and interpret.

	Panel A: VIX, C3, and Others				Panel B: Decomposed VIX, C3, and Others			
	Low	Mid	High	HmL	Low	Mid	High	HmL
Constant	0.007	-0.002	0.028***	0.021**	0.007	-0.003	0.026**	0.020**
	(0.007)	(0.007)	(0.011)	(0.009)	(0.007)	(0.007)	(0.011)	(0.009)
ΔVIX	-0.617	-0.205	-3.106*	-2.489*				
	(1.243)	(1.195)	(1.660)	(1.332)				
ΔRVOL					-0.604	-0.622	-2.747**	-2.143**
					(0.730)	(1.055)	(1.166)	(0.980)
VRP					0.652	-0.182	-6.486	-7.138*
					(3.675)	(3.718)	(5.603)	(3.969)
Cmkt	0.656***	0.792***	0.520***	-0.136	0.663***	0.795***	0.544***	-0.119
	(0.093)	(0.064)	(0.099)	(0.100)	(0.094)	(0.064)	(0.098)	(0.099)
Csize	0.088**	0.049	0.055	-0.033	0.087**	0.049	0.052	-0.035
	(0.040)	(0.039)	(0.046)	(0.033)	(0.041)	(0.039)	(0.046)	(0.032)
Cmom	-0.117	-0.136	0.073	0.190**	-0.119	-0.137	0.070	0.189**
	(0.087)	(0.092)	(0.139)	(0.094)	(0.089)	(0.094)	(0.142)	(0.096)
Cds	-0.494***	-0.203*	-0.731***	-0.237	-0.482***	-0.193	-0.697***	-0.215
	(0.135)	(0.116)	(0.192)	(0.159)	(0.142)	(0.120)	(0.200)	(0.164)
Δgpr	0.533	0.038	0.778	0.245	0.505	0.040	0.702	0.197
	(0.811)	(0.793)	(1.282)	(0.896)	(0.799)	(0.798)	(1.279)	(0.895)
$\Delta\sigma^{\text{crypto}}$	1.014**	0.488	1.440**	0.426	0.998**	0.492	1.385**	0.387
	(0.460)	(0.340)	(0.692)	(0.458)	(0.450)	(0.342)	(0.680)	(0.457)
Fiat Carry	0.349	-0.272	0.116	-0.233	0.346	-0.308	0.215	-0.131
	(0.487)	(0.441)	(0.494)	(0.460)	(0.500)	(0.458)	(0.542)	(0.484)
Illiquidity	-3.350	3.247	-5.917	-2.567	-2.884	3.073	-6.912	-4.027
	(3.811)	(4.096)	(4.583)	(3.115)	(3.955)	(4.355)	(5.278)	(3.353)
R^2	0.629	0.662	0.510	0.076	0.629	0.662	0.511	0.078

	Panel C: VIX, C5, and Others				Panel D: Decomposed VIX, C5, and Others			
	Low	Mid	High	HmL	Low	Mid	High	HmL
Constant	0.041***	0.042***	0.047***	0.005	0.042***	0.043***	0.046***	0.005
	(0.008)	(0.008)	(0.008)	(0.005)	(0.008)	(0.008)	(0.008)	(0.005)
ΔVIX	-1.110	-1.146	-2.750**	-1.640				
	(1.376)	(1.635)	(1.287)	(1.120)				
ΔRVOL					1.082	1.128	-1.483	-2.565***
					(0.840)	(1.275)	(1.018)	(0.848)
VRP					-5.493	-7.658	-5.540	-0.047
					(4.253)	(5.178)	(4.227)	(2.144)
Mkt	0.400***	0.361***	0.394***	-0.006	0.408***	0.369***	0.402***	-0.006
	(0.073)	(0.076)	(0.061)	(0.051)	(0.073)	(0.075)	(0.061)	(0.051)
Size	-0.011	-0.018	-0.009	0.003	-0.011	-0.017	-0.007	0.003
	(0.014)	(0.016)	(0.013)	(0.010)	(0.014)	(0.017)	(0.013)	(0.010)
Momentum	0.023	0.032	0.150	0.127	0.016	0.024	0.144	0.128
	(0.076)	(0.070)	(0.114)	(0.087)	(0.077)	(0.070)	(0.115)	(0.086)
Value	-0.022	-0.003	0.070	0.092***	-0.020	-0.002	0.076	0.096***
	(0.053)	(0.045)	(0.052)	(0.033)	(0.052)	(0.046)	(0.052)	(0.032)
Network	-0.009	0.036	0.154**	0.163***	-0.007	0.037	0.160**	0.167***
	(0.073)	(0.055)	(0.069)	(0.050)	(0.073)	(0.055)	(0.071)	(0.050)
Cds	-1.110***	-1.066***	-1.141***	-0.031	-1.132***	-1.090***	-1.150***	-0.018
	(0.117)	(0.117)	(0.118)	(0.072)	(0.116)	(0.113)	(0.119)	(0.070)
Δgpr	-0.037	-0.608	-0.269	-0.232	-0.098	-0.655	-0.385	-0.288
	(0.714)	(1.010)	(1.000)	(0.798)	(0.705)	(1.016)	(0.985)	(0.796)
$\Delta\sigma^{\text{crypto}}$	1.047***	0.638	1.205***	0.157	1.002***	0.598	1.151***	0.149
	(0.392)	(0.472)	(0.399)	(0.355)	(0.381)	(0.470)	(0.391)	(0.351)
Fiat Carry	-0.626	-1.462***	-0.614	0.012	-0.353	-1.157**	-0.457	-0.104
	(0.536)	(0.533)	(0.478)	(0.438)	(0.541)	(0.579)	(0.502)	(0.477)
Illiquidity	-7.723	-1.656	-9.316*	-1.592	-8.206	-2.845	-9.700	-1.494
	(5.428)	(6.696)	(5.176)	(3.468)	(6.358)	(7.366)	(5.924)	(2.844)
R^2	0.569	0.538	0.548	0.115	0.570	0.539	0.546	0.124

Table 9: Time-Series Carry Trade between Stablecoins and the U.S. Dollar

This table reports time-series carry trade between stablecoins and the U.S. dollar. The sample of USDT (December 2018) starts earlier than that of DAI (July 2020). To make a fair comparison, we also include results with the sample starting from July 2020 for both stablecoins. Carry return is daily frequency monthly return (annualized). As suggested by Cong, He and Tang (2023), we exclude the observations from the first week after the launch of each stablecoin due to the presence of abnormal price fluctuations during this period. Tracking error is calculated as $\sqrt{\frac{1}{N}\sum(S_t-1)^2}$, where N is the total number of observations for each stablecoin. S_t is the spot price of the stablecoin on day t. A higher tracking error indicates a higher level of depegging risk. Newey-West standard errors with four lags are reported in parentheses. *, **, and *** represent statistical significance at the 10%, 5%, and 1% levels, respectively. Data span from July 2020 to November 2021 for DAI and from December 2018 to November 2021 for USDT.

	DAI	USDT
Since Inception		
Average carry return	10.92%***	9.28%***
	(3.54%)	(0.76%)
Return on interest rate differential	11.70%***	9.48%***
	(3.78%)	(0.70%)
Return on change in prices	-0.78%	-0.20%
	(0.71%)	(0.40%)
Std	7.09%	2.62%
Sharpe ratio	1.54	3.54
Tracking error	0.008	0.005
Since July 2020		
Average carry return	10.92%***	9.98%***
	(3.54%)	(0.73%)
Return on interest rate differential	11.70%***	9.96%***
	(3.78%)	(0.73%)
Return on change in prices	-0.78%	0.01%
	(0.71%)	(0.08%)
Std	7.09%	1.51%
Sharpe ratio	1.54	6.62
Tracking error	0.008	0.001

Appendix

A Staking Reward Rate and Interest Rate of Cryptocurrency

We define the carry of a cryptocurrency as the interest rate differential between the cryptocurrency and the U.S. dollar. On the Bitfinex platform, investors can lend their cryptocurrency to others to earn interest. The interest rate on this platform is influenced by the supply and demand for borrowing and lending cryptocurrencies. Borrowers use the borrowed cryptocurrency for margin trading. For instance, if they anticipate that the price of Bitcoin will decrease relative to the U.S. dollar, they can open a short position by borrowing Bitcoin from others on the platform and paying interest at a specified rate. Loan durations range from 2 to 120 days. The platform charges a funding fee of 15% of the interest (or 18% if investors opt to hide their order). Consequently, there is a difference between the interest rate earned by lenders and the rate paid by borrowers.

The interest rate of cryptocurrency we utilize is different from the staking reward rate of cryptocurrency. Staking rewards are earned when investors stake their cryptocurrency because it helps validate transactions and improve network security (Chitra, 2021). This mechanism is distinct from the loanable programs of cryptocurrency that we study. Cong, He and Tang (2023) focus on the carry trade of stakable tokens. Our paper is a complementary study to their research.

B Relationship between Fiat Currency Carry Trade and Cryptocurrency Carry Trade

The carry trade in the fiat currency market is typically implemented with foreign exchange futures. Koijen, Moskowitz, Pedersen and Vrugt (2018) define carry of a currency as $\frac{S_t - F_t}{F_t}$, where S_t is the spot price of the currency and F_t is the futures price, both quoted per U.S. dollar. If covered interest rate parity holds, $\frac{S_t - F_t}{F_t} = \frac{i_t^{f^*} - i_t^{USD}}{1 + i_t^{USD}}$. $i_t^{f^*}$ is the interest rate of the currency and i_t^{USD} is the interest rate of the U.S. dollar. Therefore, our definition of cryptocurrency carry is consistent with this definition of fiat currency carry in Koijen, Moskowitz, Pedersen and Vrugt (2018) as the interest rate of cryptocurrencies corresponds to that of fiat currencies. As only a limited number of cryptocurrencies have futures contracts, we implement carry trade of cryptocurrency in the spot market by borrowing cryptocurrencies with lower interest rates and depositing those with higher interest rates. Specifically, we first sort all cryptocurrencies in our sample into three equal-weighted portfolios based on their carry. Portfolio 3 has the highest-carry cryptocurrencies, while portfolio 1 has the lowest-carry cryptocurrencies. Each portfolio is financed using U.S. dollar loans, and the cryptocurrencies are deposited on the Bitfinex platform to earn interest. Therefore, the log return of cryptocurrencies in each portfolio is calculated as

$$r_t = i_{t-1}^c - i_{t-1}^{USD} + s_t - s_{t-1}, \tag{13}$$

where i^c_{t-1} and i^{USD}_{t-1} are interest rate of the cryptocurrency and the U.S. dollar at time $t-1$, respectively. s_t is the log price of cryptocurrency at time t. This is consistent with the calculation of excess return of fiat currency in Lettau, Maggiori and Weber (2014). The price of cryptocurrency denoted by the U.S. dollar is an analogy of the exchange rate of fiat currency in their paper. To construct the carry trade portfolio, we long portfolio 3 and short portfolio 1.[22] The carry return is the return of portfolio 3 minus the return of portfolio 1. We rebalance the portfolios at weekly frequency.

C Difference between Borrowing and Lending Rates

In this subsection, we explicitly take into account the difference between borrowing and lending rates. If a cryptocurrency is in the long position,

$$r_{t+1} = i^{c-dep}_t - i^{USD}_t + \Delta s_{t+1}\,, \tag{14}$$

and if a cryptocurrency is in the short position,

$$r_{t+1} = i^{USD}_t - i^{c-borw}_t - \Delta s_{t+1}\,, \tag{15}$$

where i^{c-dep}_t and i^{c-borw}_t are the interest rate of depositing and borrowing the cryptocurrency, respectively. s_t is log price of the cryptocurrency in U.S. dollars. Since the carry strategy shorts the Low portfolio and longs the High portfolio, we recalculate Low portfolio return using Equation (15) and update the high-minus-low carry return accordingly. The borrowing cost-adjusted cryptocurrency carry portfolio performance is reported in Table A1.

Overall, the portfolio return is qualitatively similar to the HmL portfolio in Table 4, suggesting that differences in borrowing and lending rates do not significantly influence cryptocurrency carry returns. This is because cryptocurrency carry returns are predominately driven by price fluctuations of cryptocurrencies instead of interest rate differentials.

[Insert Table A1 Here]

[22] Short portfolio 1 refers to the trading that borrow these cryptocurrencies and sell them in the spot market for the U.S. dollar, and then deposit the U.S. dollar to earn risk-free interest rate.

Table A1: Summary Statistics of Borrowing-Cost-Adjusted Carry Trade Portfolios

This table presents summary statistics of carry portfolios when we consider the difference between borrowing and lending interest rates. The lending rate (for the investors who lend their cryptocurrencies to others) of a cryptocurrency is equal to 0.85 multiplied by the borrowing rate (for the investors who borrow cryptocurrencies from others) of the cryptocurrency. The reason is that the fund providers have to pay a fee equal to 15% of the interest they earned to the platform (if the funding is not opened with a hidden offer). Short Position (Long Position) represents the portfolio containing tokens with the lowest (highest) interest rates. Long-Short represents the long-short portfolio that buys tokens in the Long Position portfolio and shorts those in the Short Position portfolio. Portfolios are rebalanced weekly. Average return, return on Δs, return on interest differential, and corresponding standard errors are annualized and in percentage points. We multiply the weekly mean by 52 and weekly standard deviation (volatility) by $\sqrt{52}$ to calculate the annualized mean and standard deviation. Newey-West standard errors with four lags are reported in parentheses. *, **, and *** represent statistical significance at the 10%, 5%, and 1% levels, respectively. The sample period is October 2016 to November 2021.

	Short Position	Long Position	Long-Short
Average return	-50.6	93.9*	43.3*
	(48.9)	(54.4)	(24.0)
Return on interest rate differential	0.2	19.1***	19.3***
	(0.2)	(1.3)	(1.3)
Return on change in prices	-50.7	74.7	24.0
	(48.9)	(54.3)	(24.1)
Vol	93.8	102.7	58.9
Skewness	-0.15	0.63	1.95
Kurtosis	4.3	8.3	21.1
Sharpe ratio	-0.54	0.91	0.74
# obs.	268	268	268